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(54) **SMART RING**

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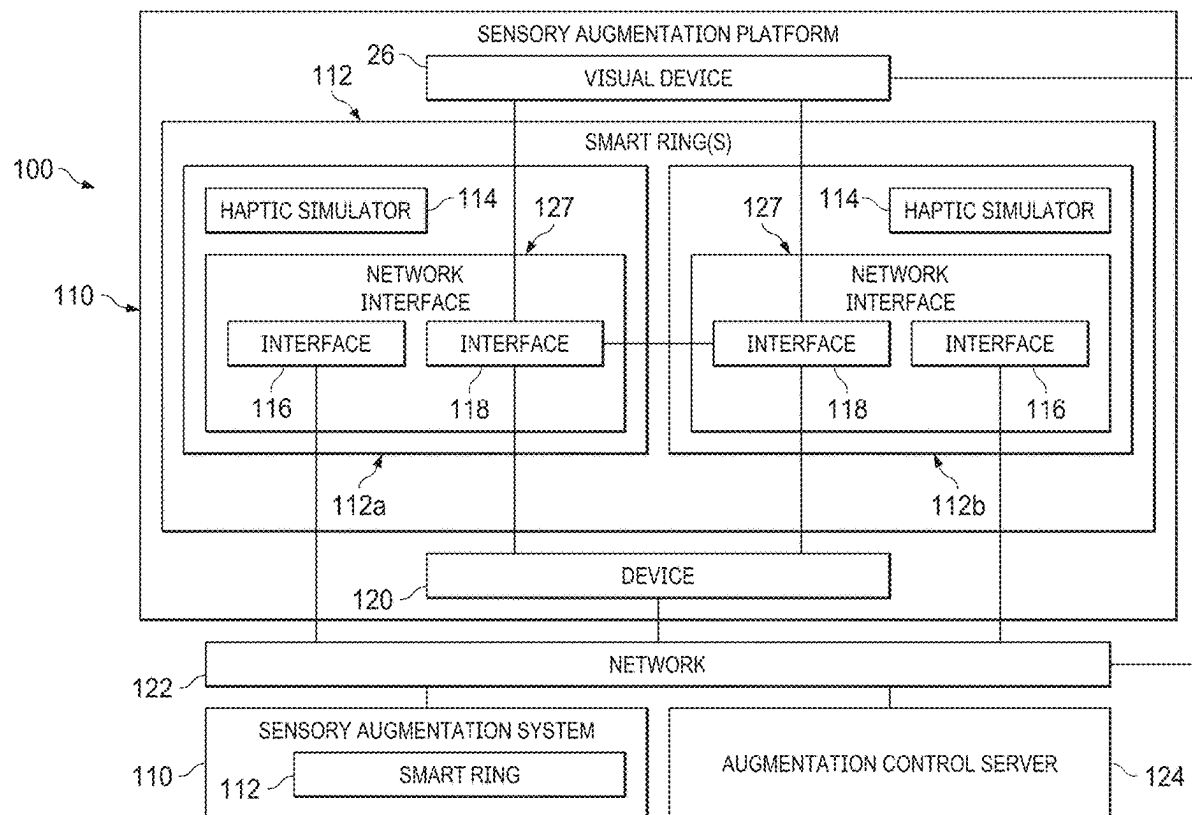
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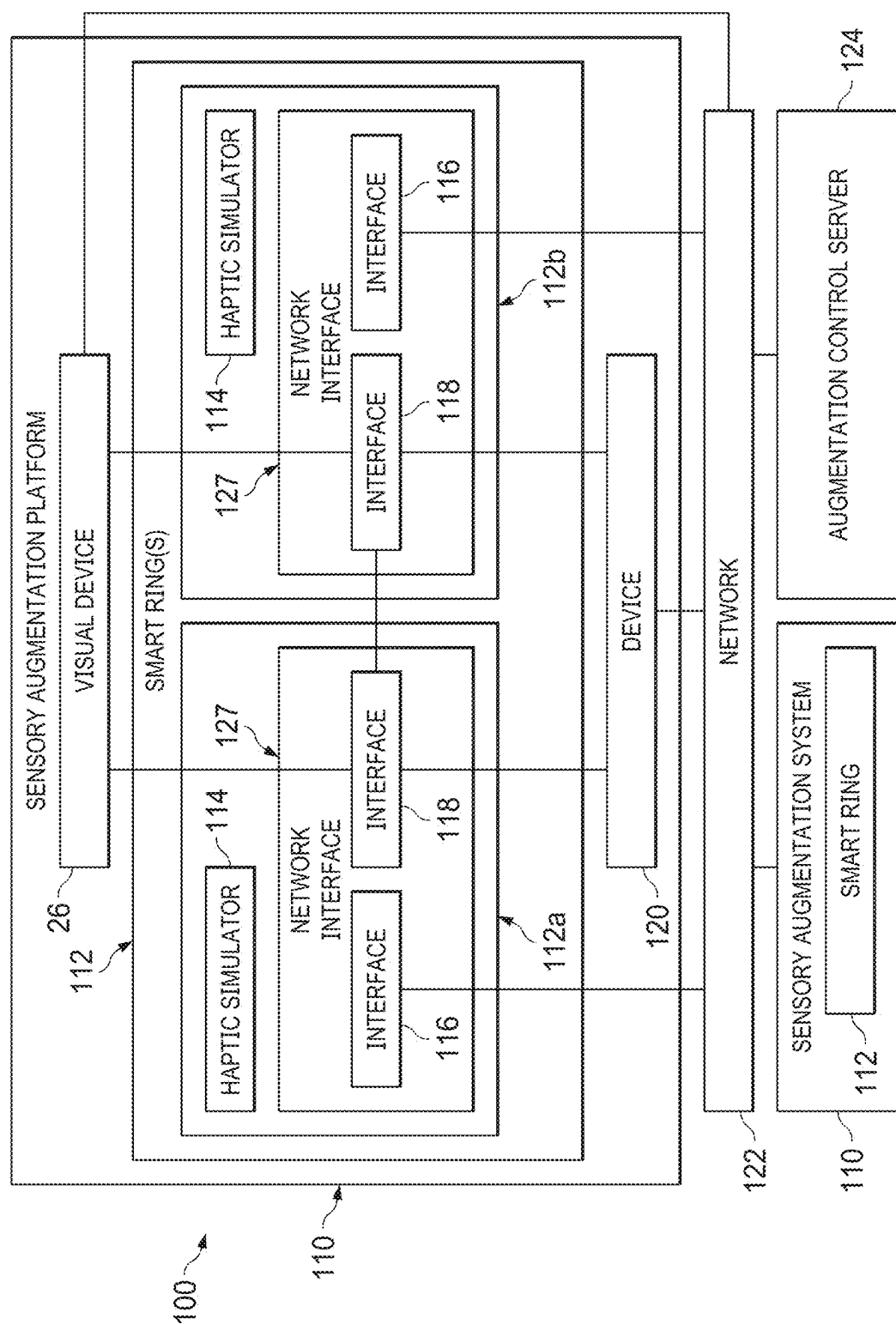
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(57)

ABSTRACT

In an example, a method includes receiving a data input. The method also includes mapping the data input to a temporal vibration pattern from among multiple temporal vibration patterns. The method also includes controlling a haptic stimulator to provide haptic stimulation according to the temporal vibration pattern.





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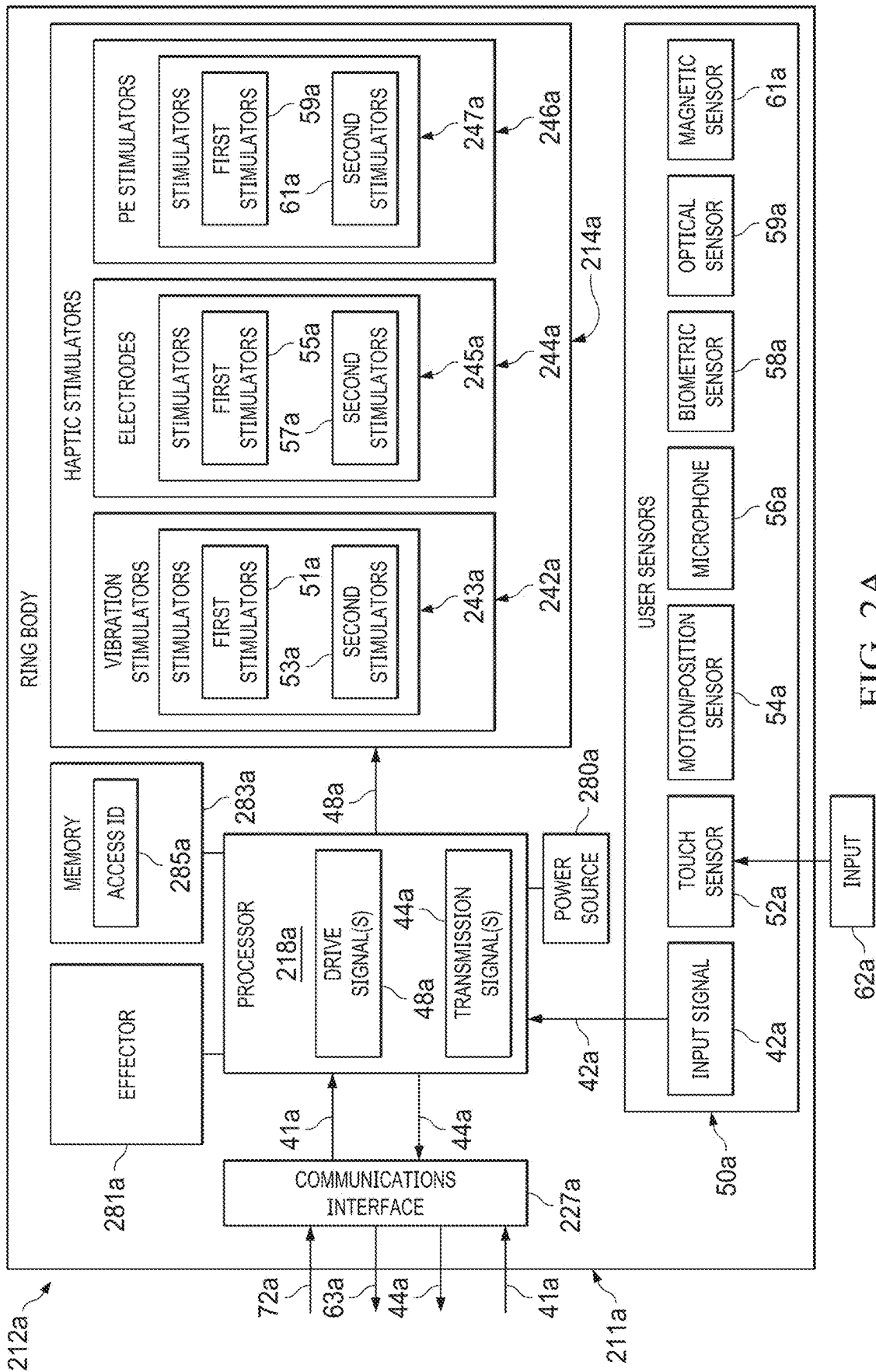
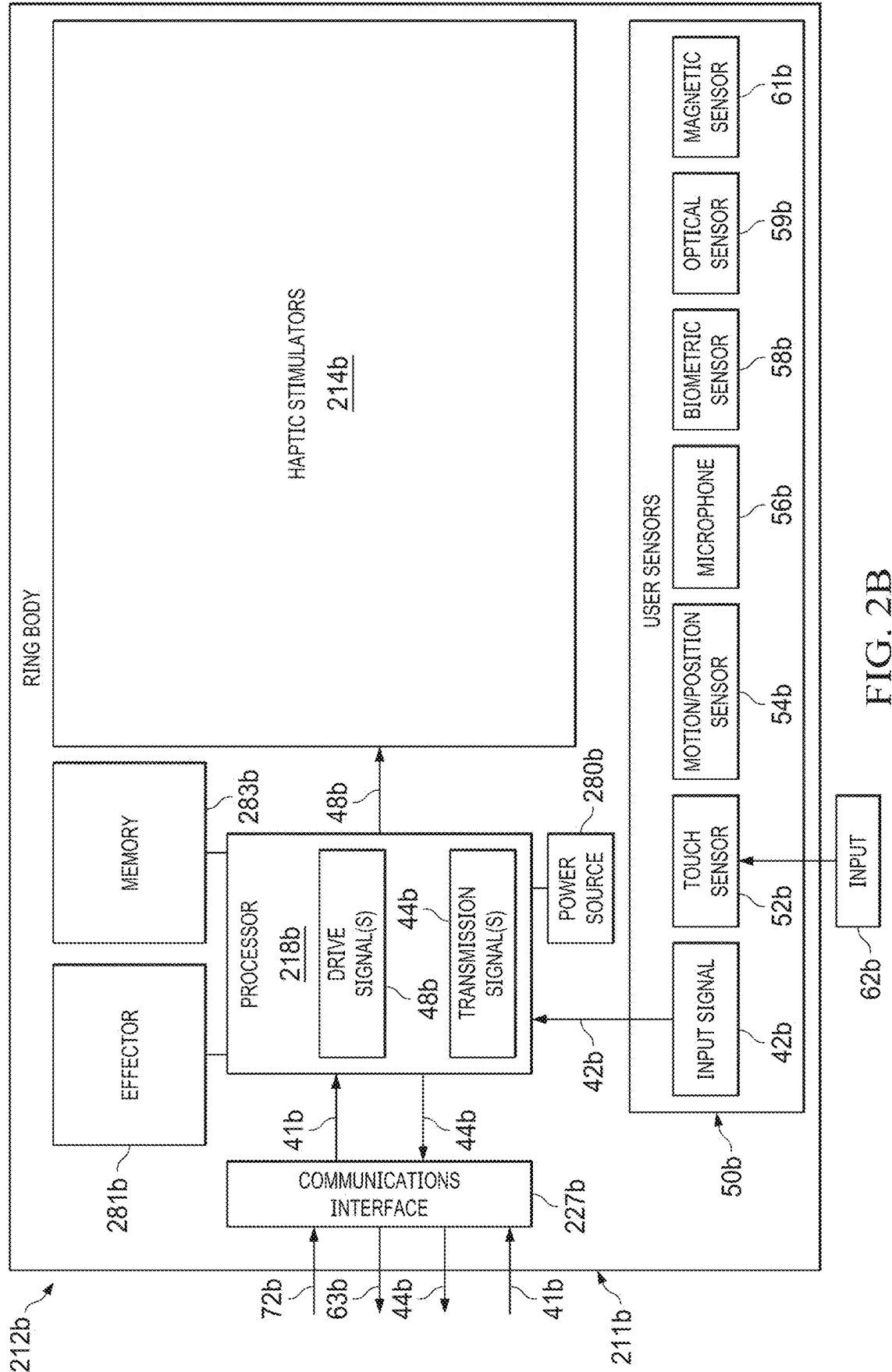


FIG. 2A



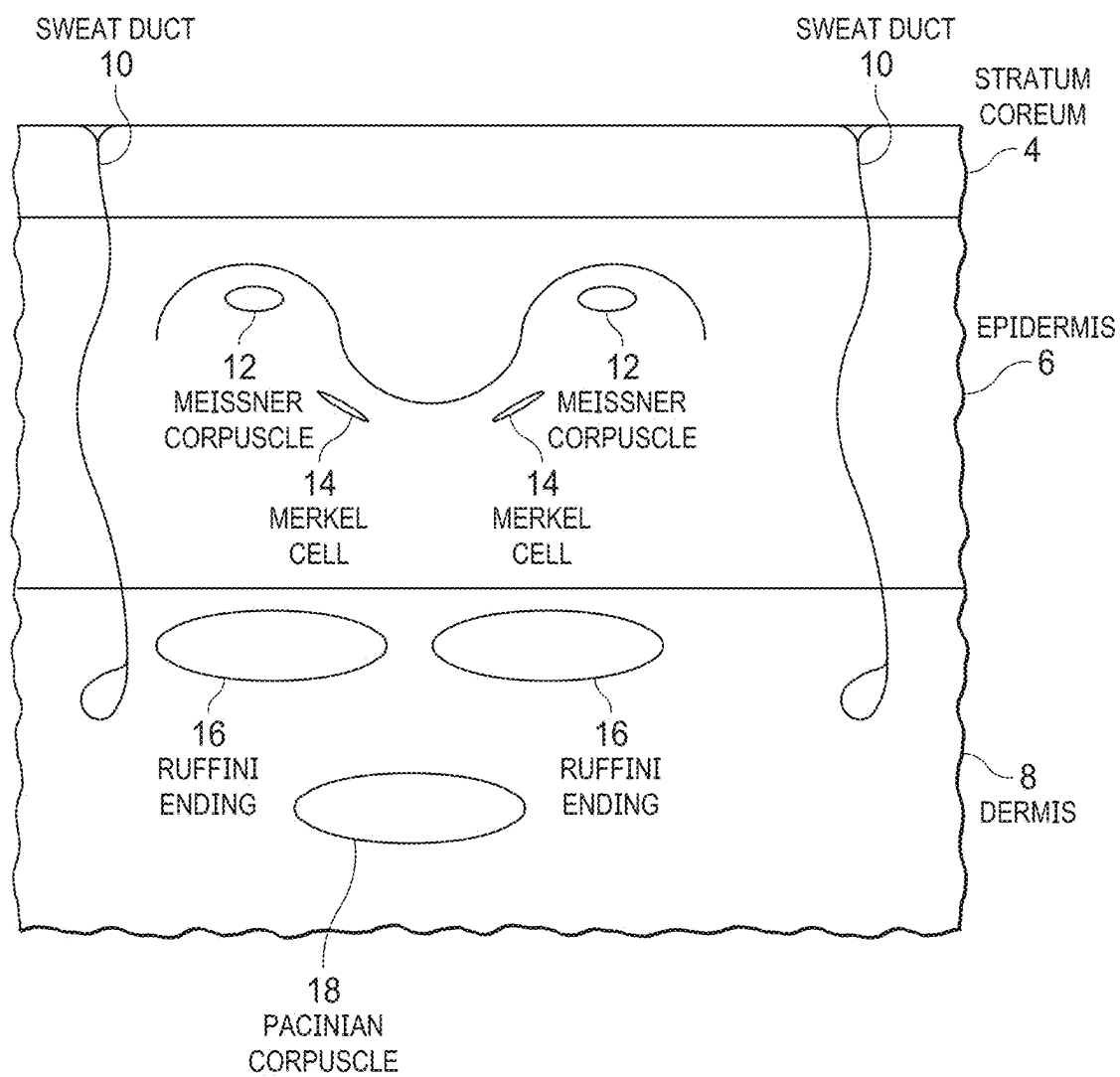


FIG. 3

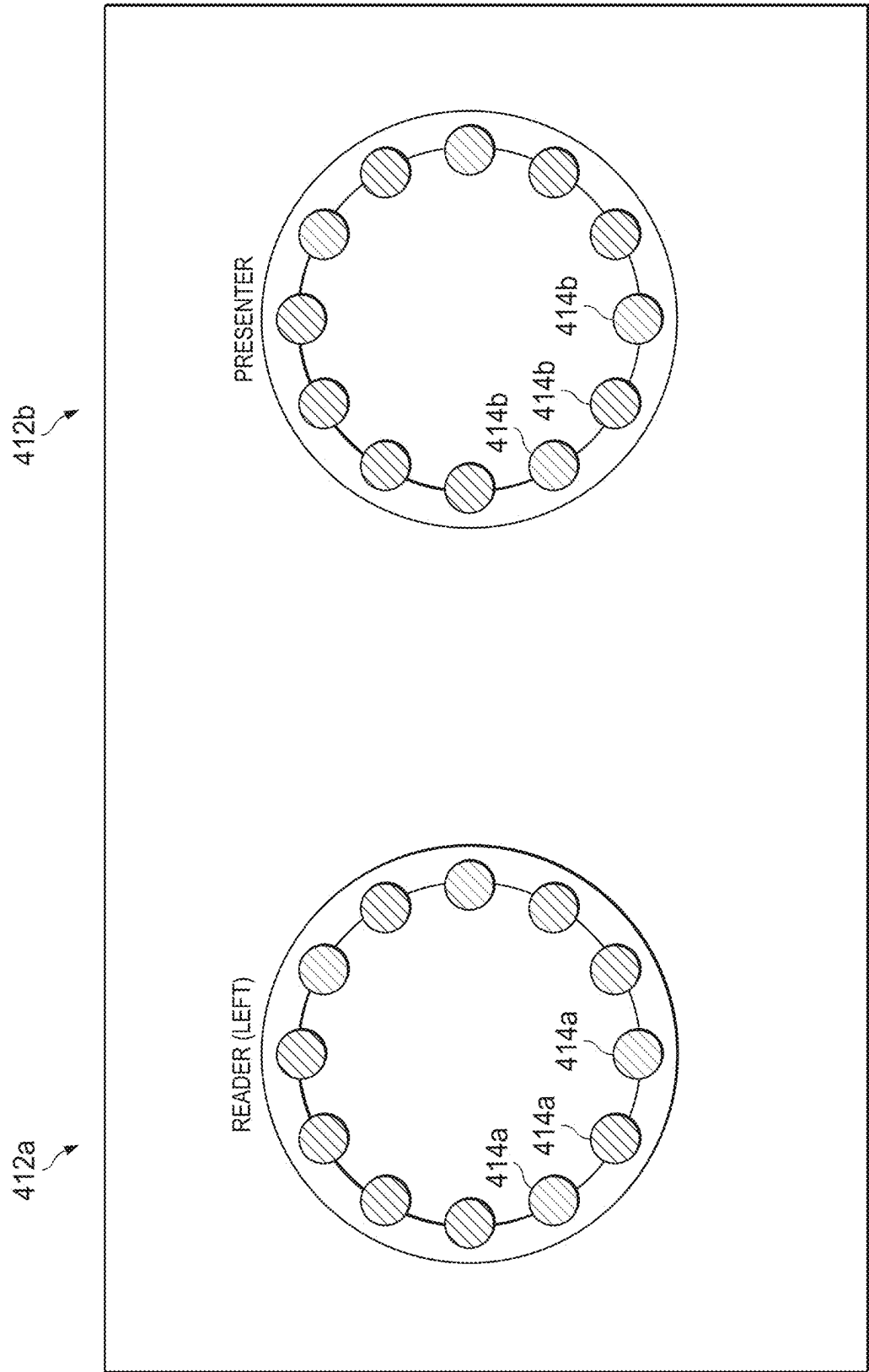


FIG. 4A

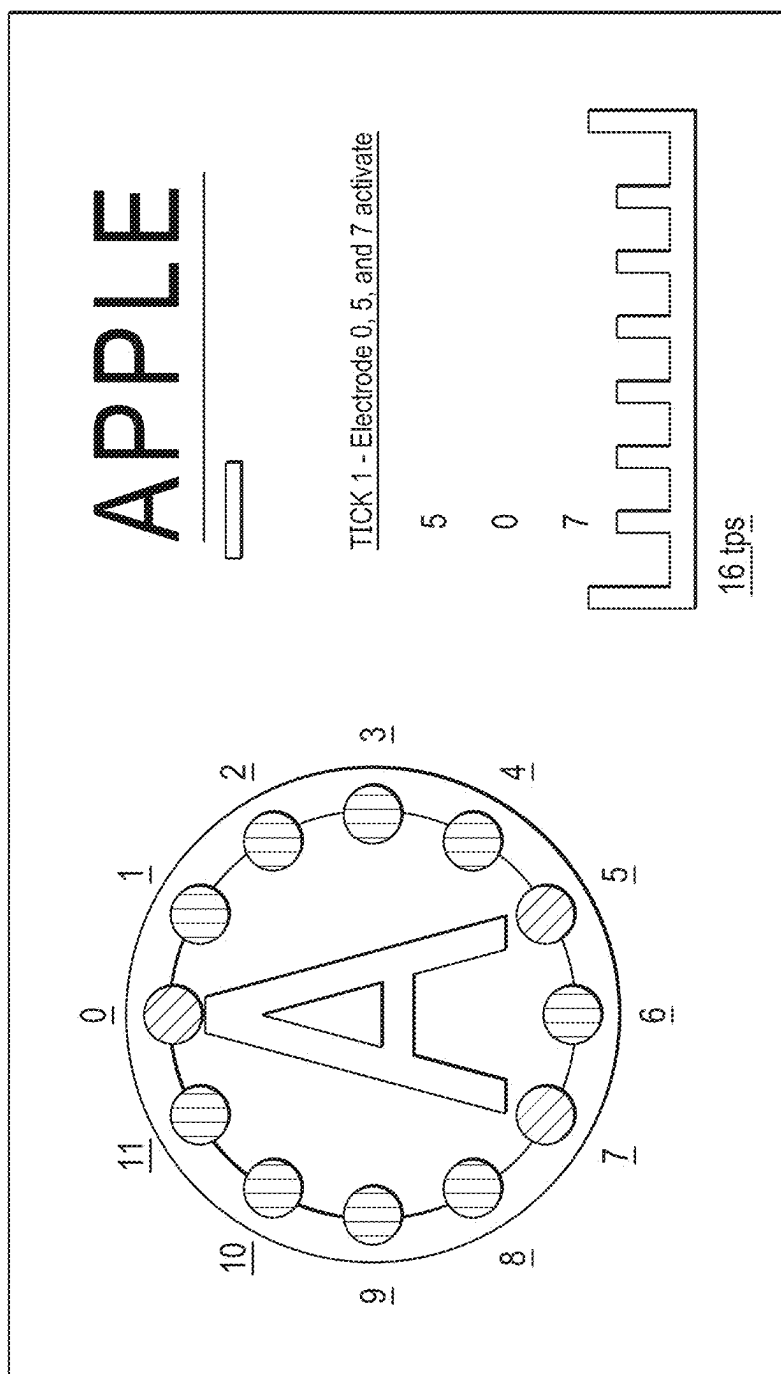


FIG. 4B

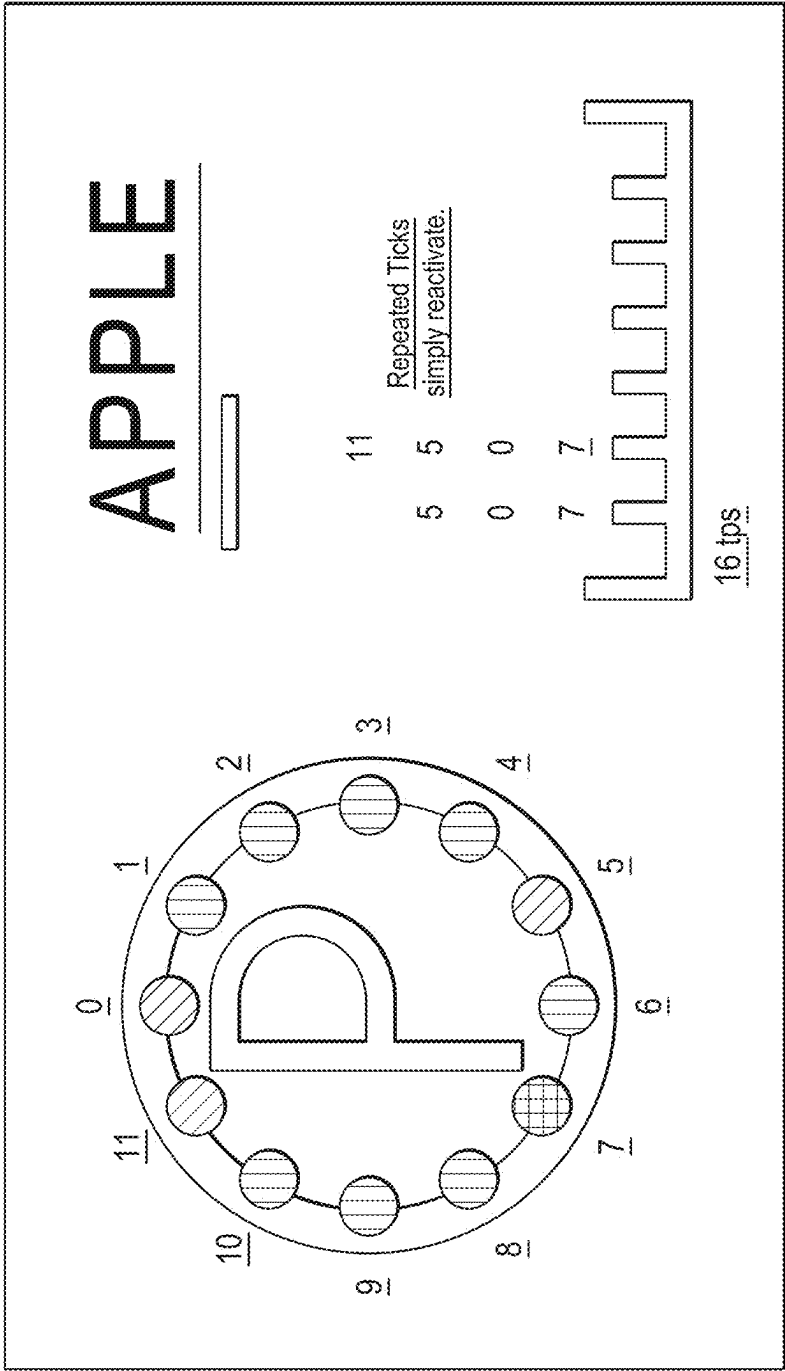


FIG. 4C

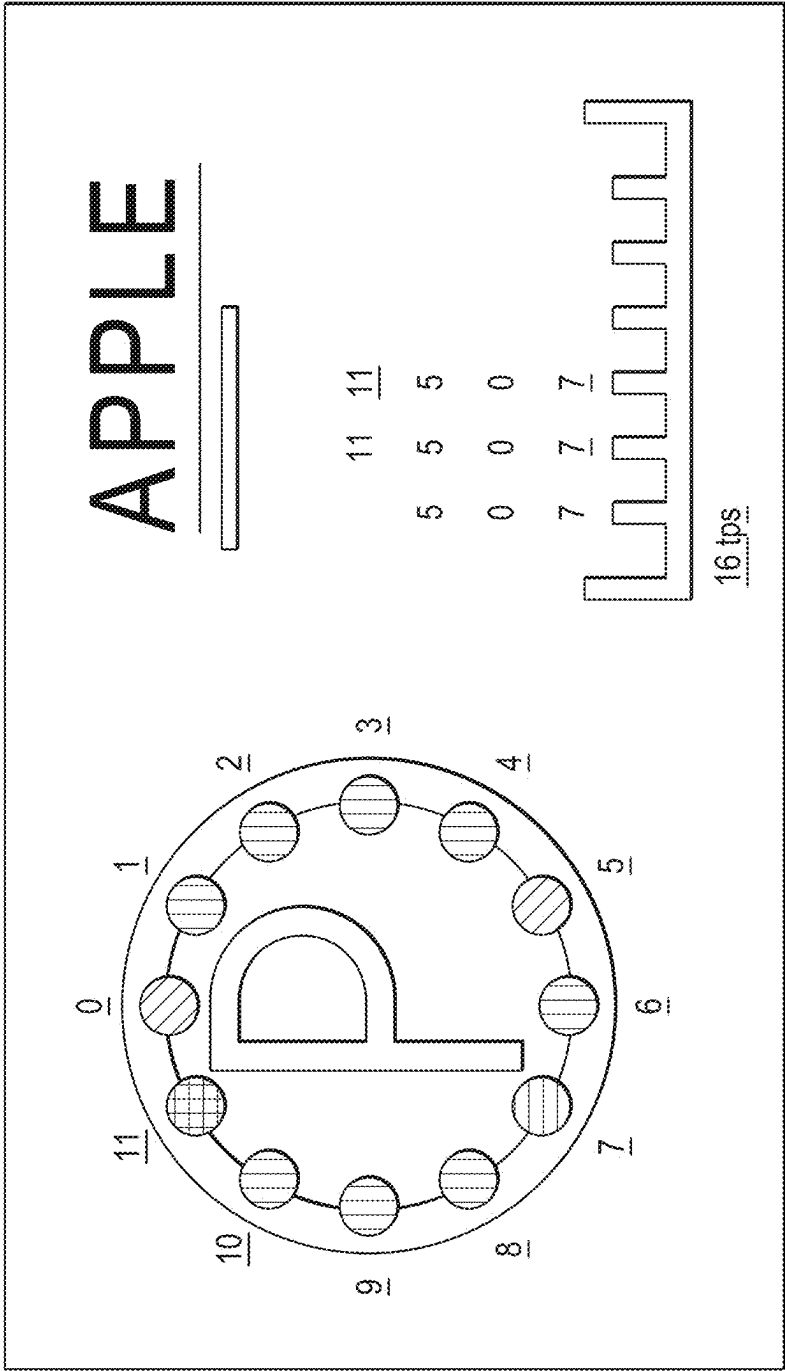


FIG. 4D

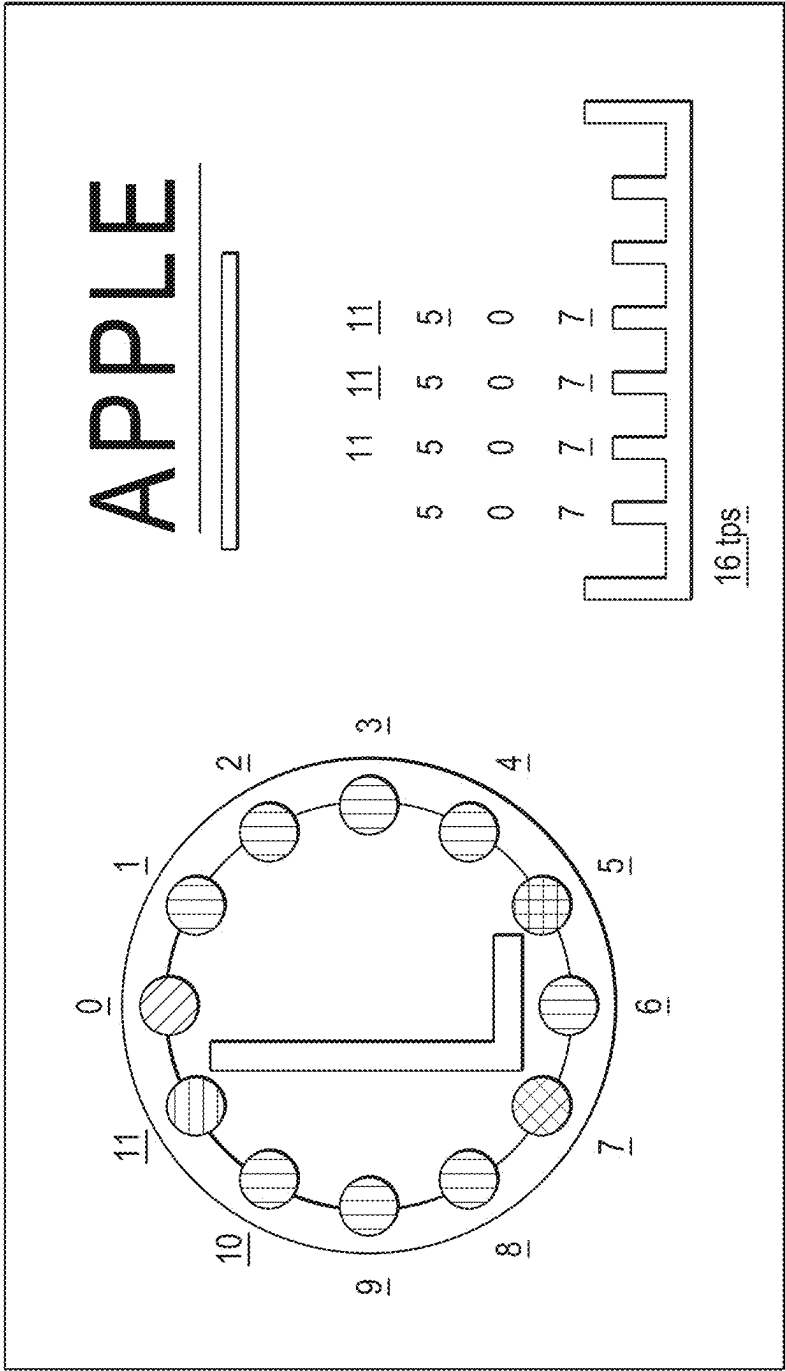


FIG. 4E

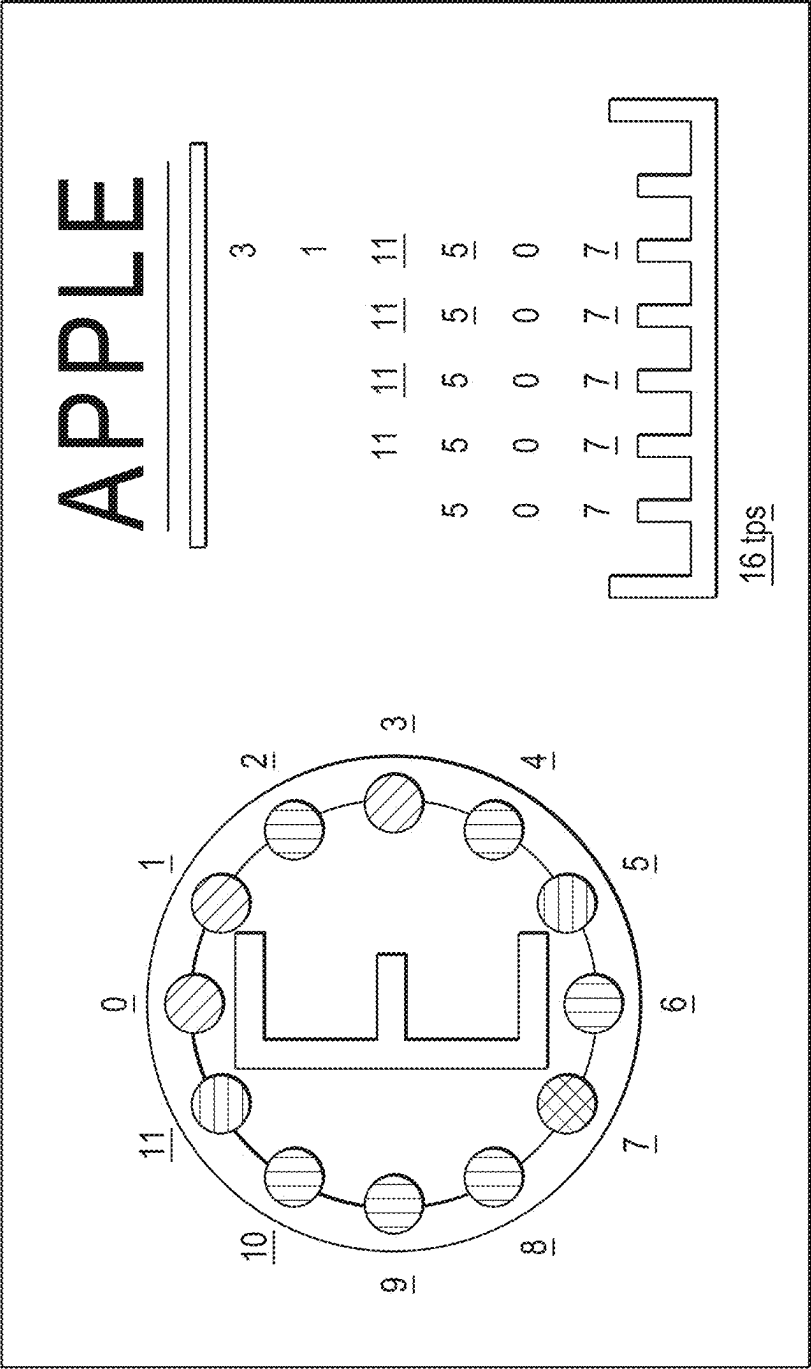


FIG. 4F

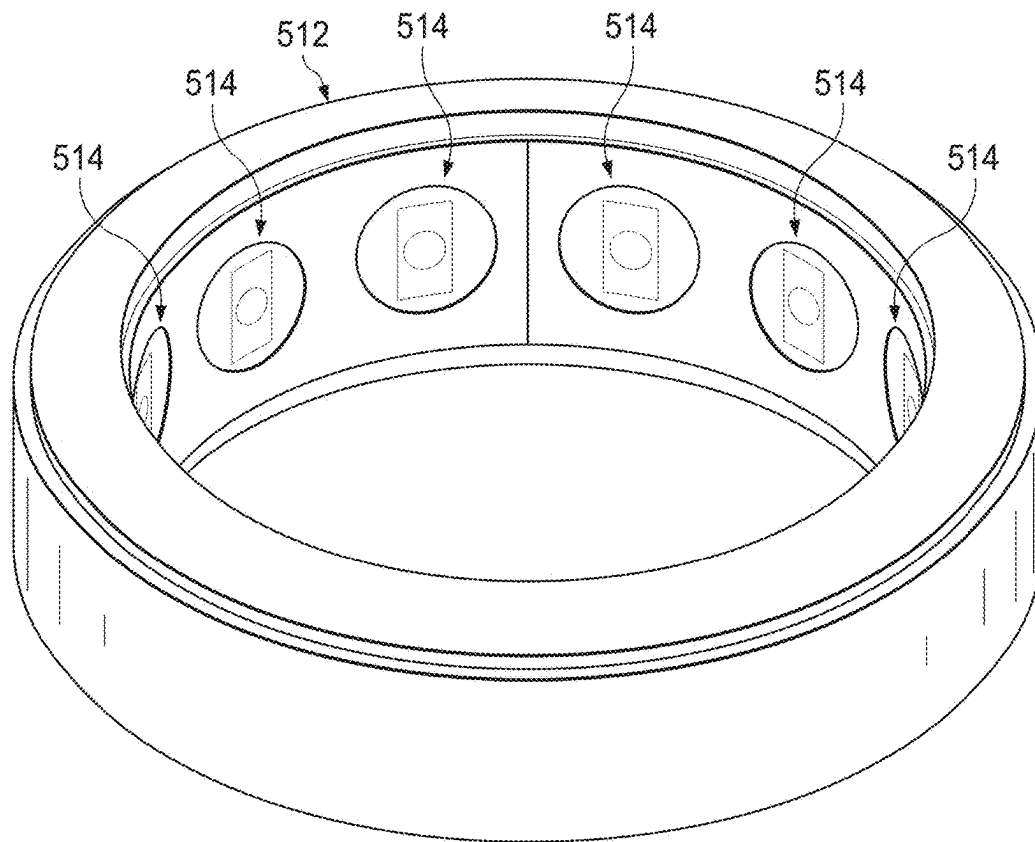


FIG. 5

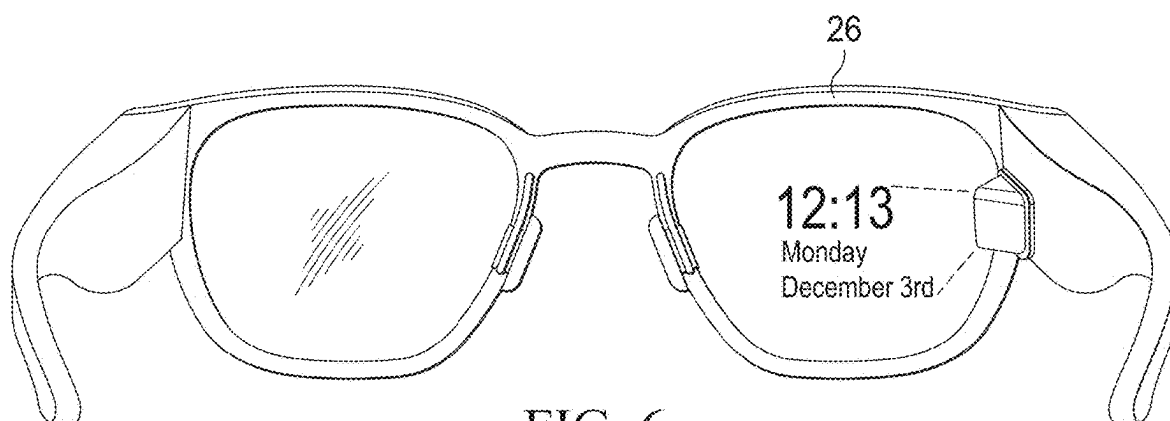


FIG. 6

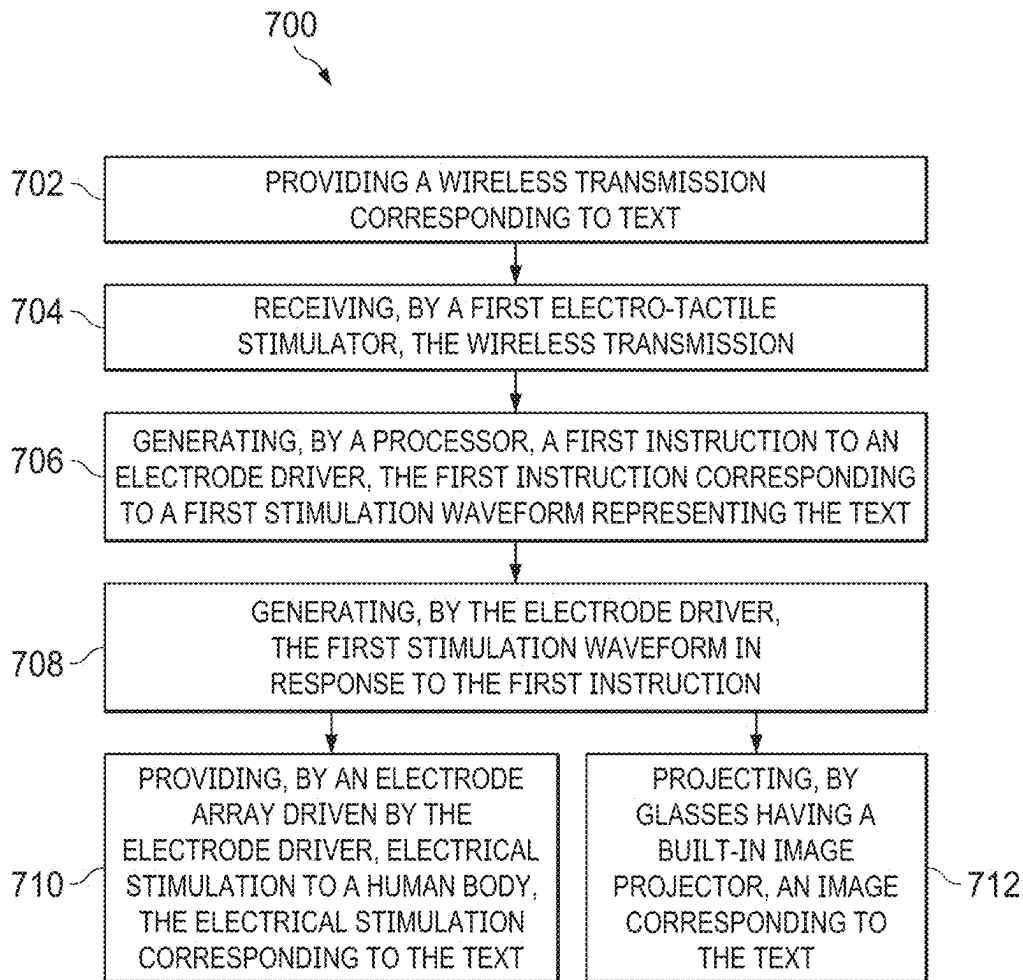


FIG. 7

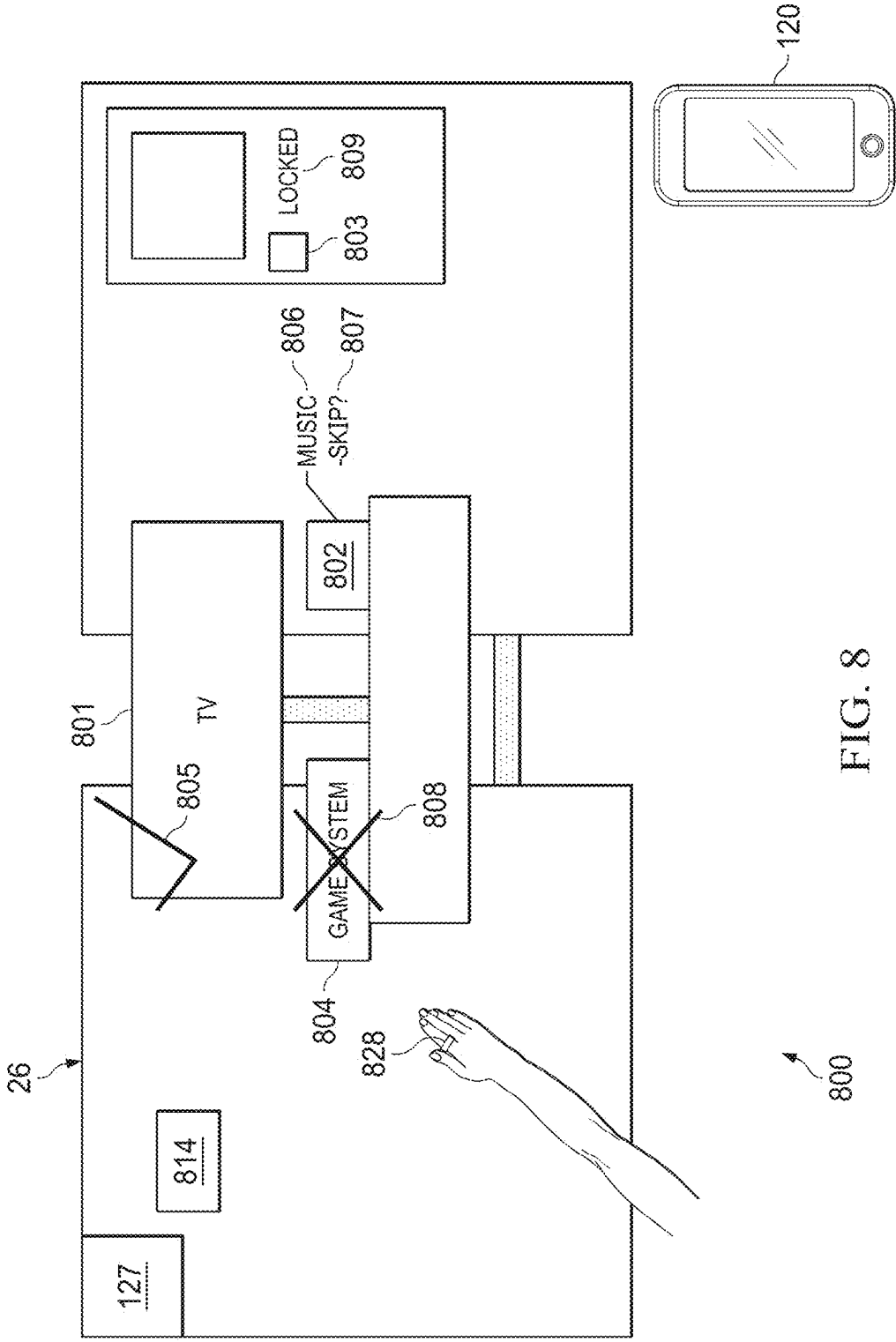


FIG. 8

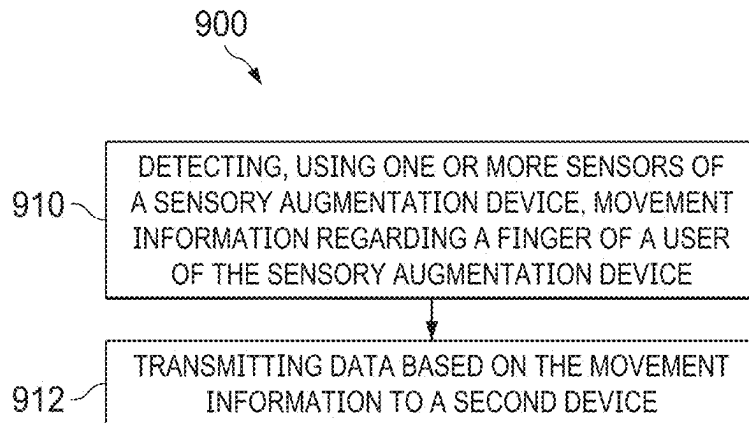


FIG. 9

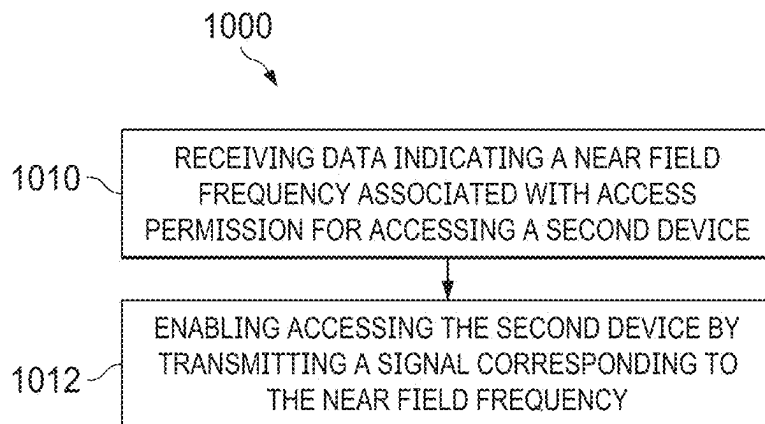


FIG. 10

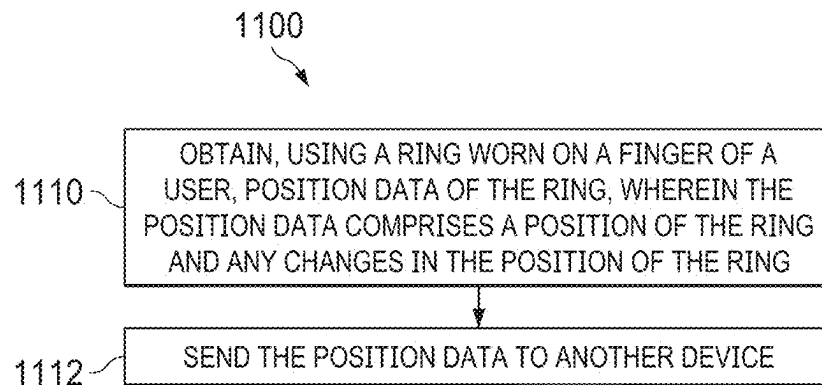


FIG. 11

SMART RING

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 18/347,384, filed on Jul. 5, 2023 by Landon Nickerson entitled, “Smart Ring” which is a continuation in part of U.S. patent application Ser. No. 16/944,506, filed on Jul. 31, 2020 by Landon Nickerson entitled, “Smart Ring,” now U.S. Pat. No. 12,008,161, which claims priority to U.S. Provisional Patent Application No. 62/905,239, filed on Sep. 24, 2019, and U.S. Provisional Patent Application No. 62/990,059, filed on Mar. 16, 2020, all of which are incorporated by reference herein in their entirety.

BACKGROUND

[0002] The present disclosure relates generally to a system and method for sensory augmentation via a smart ring.

[0003] Frequently, humans desire to communicate with others at a distance or with machines. However, typical sensory data provided to a human by a machine-human interface occupies one of the five senses, especially hearing or sight, rendering that sense distracted from other inputs. Moreover, individuals with disabilities such as hearing loss or vision impairment have difficulty interacting with such machine-human interfaces. Existing machine-human interfaces are often unsuitable for high-distraction environments, environments necessitating silences, or environments where a user must actively monitor real-world surroundings rather than an electronic device.

[0004] Additionally, existing smart accessory devices, such as watches or wrist-worn devices, are unsuitable for detecting hand movements of a user because they are unable to detect finger movement. Additionally, other approaches using watches or wrist-worn devices are unsuitable for communicating information using haptic stimulation for anatomical reasons related to lack of nerves or particular nerves sufficiently close to the skin in the wrist area.

SUMMARY

[0005] In an example, a smart ring includes a ring body configured to be worn on a finger of a user. The smart ring also includes at least one haptic stimulator configured to receive a drive signal and communicate data to the user via a haptic stimulation of the finger. The smart ring also includes a processor coupled to the at least one haptic stimulator. The at least one processor is configured to receive a notification. The at least one processor is also configured to generate the drive signal based on the notification, wherein the drive signal has a unique correspondence to a characteristic of the notification and corresponds to a temporal vibration pattern. The at least one processor is also configured to drive the at least one haptic stimulator according to the drive signal to communicate the data to the user, the data being representative of the characteristic.

[0006] In some examples, a smart ring includes a ring body configured to be worn on a finger of a user. The smart ring also includes a touch sensor coupled to the ring body, wherein the touch sensor is configured to receive a touch input from the user and convert the touch input into an input signal. The smart ring also includes a motion sensor coupled to the ring body, wherein the motion sensor is configured to provide position data representative of a position of the

smart ring in a Cartesian coordinate system. The smart ring also includes at least one processor coupled to the capacitive touch sensor. The at least one processor is configured to receive the input signal. The at least one processor is configured to, responsive to receiving the input signal, activate the motion sensor and obtain the position data. The at least one processor is configured to form a transmission signal based on the position data. The smart ring also includes a communications interface coupled to the at least one processor. The communications interface is configured to transmit the transmission signal to another device.

[0007] In some examples, a method includes receiving a data input. The method also includes mapping the data input to a temporal vibration pattern from among multiple temporal vibration patterns. The method also includes controlling a haptic stimulator to provide haptic stimulation according to the temporal vibration pattern.

[0008] In some examples, a computer-program product comprising computer-executable instructions stored on at least one non-transitory computer medium. The instructions are executable by at least one processor to cause the at least one processor to receive a notification. The instructions are executable by at least one processor to further cause the at least one processor to generate a drive signal based on the notification. The drive signal corresponds to a vibration pattern and has a one-to-one correspondence to a characteristic of the notification. The instructions are executable by at least one processor to further cause the at least one processor to drive at least one haptic stimulator according to the drive signal to convey the data to the user, the data being representative of the characteristic.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] For a more complete understanding of various embodiments of the present disclosure and the advantages thereof, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, wherein like reference numerals represent like parts, and in which:

[0010] FIG. 1 illustrates an example of a sensory augmentation platform to communicate using haptic stimulation, in accordance with various embodiments;

[0011] FIG. 2A illustrates an example block diagram of a smart ring, in connection with various embodiments;

[0012] FIG. 2B illustrates an example block diagram of a smart ring, in connection with various embodiments;

[0013] FIG. 3 illustrates an example depiction of layers of skin and human nerves, in accordance with various embodiments;

[0014] FIG. 4A shows an example of a smart ring, such as the smart ring of FIGS. 1 and 2A and 2B, and that includes haptic stimulators, in accordance with various embodiments;

[0015] FIGS. 4B-4F depict an example schema for communicating a multi-character English word to a user by activating different haptic stimulators of the smart ring of FIG. 4A, in accordance with various embodiments;

[0016] FIG. 5 illustrates an example of a smart ring including example haptic stimulators, in accordance with various embodiments;

[0017] FIG. 6 illustrates an example visual device operable in connection with a smart ring to provide visual information, in accordance with various embodiments;

[0018] FIG. 7 illustrates a flowchart of an example method of sensory augmentation, in accordance with various embodiments;

[0019] FIG. 8 illustrates a use scenario combining multiple aspects of the sensory augmentation platform of FIG. 1, in accordance with various embodiments;

[0020] FIG. 9 illustrates a flowchart of an example method of sensory augmentation using one or more rings, such as one or more of the rings of FIG. 1, 2A, 2B, 4A-F, or 5;

[0021] FIG. 10 illustrates a flowchart of an example method of sensory augmentation using one or more rings, such as one or more of the rings of FIG. 1, 2A, 2B, 4A-F, or 5; and

[0022] FIG. 11 illustrates a flowchart of an example method of sensory augmentation using one or more rings, such as one or more of the rings of FIG. 1, 2A, 2B, 4A-F, or 5.

DETAILED DESCRIPTION

[0023] The present disclosure is generally described in detail with reference to embodiments illustrated in the drawings. However, other embodiments may be used and/or other changes may be made without departing from the spirit or scope of the present disclosure. The illustrative embodiments described in the detailed description are not meant to be limiting of the subject matter presented herein.

[0024] Reference will now be made to the exemplary embodiments illustrated in the drawings, and specific language will be used to describe the same. It will nevertheless be understood that no limitation of the scope of the invention is thereby intended. Alterations and further modifications of the features illustrated and/or described herein, and additional applications of the principles of the embodiments as illustrated herein, which would occur to one skilled in the relevant art in view of this disclosure, are to be considered within the scope of the invention.

[0025] Some embodiments include a sensory augmentation device or system configured to communicate information to a user using tactile sensations, such as haptic sensations, electrical stimulation, and/or pressure sensations. A user may learn to associate meaning with haptic sensations, thereby facilitating communication between a device and a user. Some embodiments of systems, devices, and/or methods disclosed herein may provide a sequence of such sensations in parallel and/or in series over time. Moreover, a user may be readily trained to recognize the sequence of such sensations as corresponding to meanings. In various embodiments, a wearable headpiece, such as a pair of glasses with a built-in visual image projector may be worn and visual images may be projected that correspond to different haptic stimuli. The user may thus be trained to understand the meaning of the different haptic stimuli by being shown relevant visual images. A user may also enjoy training through use of a smartphone application, or a computer program, or an online training portal. For example, a computer application or other program may be loaded on a computer and screen displays may correspond to different haptic stimuli. Furthermore, augmented reality, virtual reality, and other such immersive environments may be created by the combination of a visual stimuli and haptic stimuli. Still furthermore, bone conduction headphones or other aural stimulus devices may be incorporated into the pair of glasses to further provide auditory stimuli in connection with the haptic stimuli. In this manner, stimulation

of vision, sound, and haptic senses may be combined to communicate information or simulate seen, heard, and felt experiences.

[0026] Many example use cases are contemplated. In one example embodiment, an application is installed on a handheld computing device such as a smartphone. The application may connect the smartphone to the sensory augmentation device providing the haptic stimulation. For instance, text messages, alarms, and other alerts that would generate a visual or audible alert at the handheld computing device may generate a haptic alert via the systems, devices, and methods disclosed herein. An application may also be run on a desktop or laptop computing device. The application may connect the computer to the sensory augmentation device providing the haptic stimulation. For instance, training scenarios, games, and other visual displays on the computer may correspond to haptic alerts. In this manner, a user may be trained to understand the meaning of different alerts in an immersive electronic training environment, through scenario based training episodes, and the like.

[0027] Moreover, multiple sensory augmentation devices may provide haptic information to multiple body areas. For instance, multiple finger rings, toe rings, bracelets, and/or other devices may be used. The different devices may play different communicative roles. For example, a ring may be placed on each of a first finger and a second finger. Upon receipt of a message, both rings may stimulate a corresponding finger to indicate the presence of an alert, subsequently one ring may provide haptic stimulation to communicate the nature or origin of the alert, and another ring may provide haptic stimulation to communicate content of the alert.

[0028] Systems, devices, and methods to communicate via such haptic stimuli may also include aspects to receive user input. For instance, a ring that provides haptic stimulation may also include a touch sensor that receives inputs. For instance, human input to the sensory augmentation devices, systems, and/or methods may include swipes, taps, multi-taps, deep taps, voice dictation, motion or gesture tracking (e.g., via gyroscopic, piezoelectric, optical, or other sensors). In some examples, the ring includes active and inactive operational states. In the inactive operational state, at least some human input to the ring is ignored. In the active operational state, at least some human input to the ring ignored during the inactive operational state is not ignored. To toggle between the inactive state and the active state, in some examples, human input is received. For example, second human input received while first human input is also received, may be processed. Conversely, second human input received while first human input is not received, may be ignored. In some examples, the first human input may include swipes, taps, multi-taps, deep taps, voice dictation, and the second human input may include motion or gesture tracking (e.g., via gyroscopic, piezoelectric, optical, or other sensors).

[0029] Users thus may readily translate conversations with foreign language speakers, receive and understand navigation instructions from a mapping device without looking at the device or otherwise becoming visually distracted, may control devices, may learn languages, may control game systems, may engage in virtual reality environments, and may conduct financial transactions, among other uses, without need for engaging vision or hearing senses.

[0030] With reference now to FIG. 1, an example of a sensory augmentation platform 100 that includes one or

more smart rings **112** of a first user and one or more smart rings **112** of another user or users is disclosed. The sensory augmentation platform **100** may include a sensory augmentation system **110** of a user, an augmentation control server **124** (e.g., a spatially localized computing device also referred to herein as a hub or smart hub), and one or more sensory augmentation systems **110** of other users. Each of the sensory augmentation systems **110** may include the one or more smart rings **112**, a computing device **120** (e.g., a smartphone), and/or a visual device **26**. In some examples, the sensory augmentation platform **100** is configured to communicate human readable data to a user via haptic stimulation using the one or more smart rings **112**. In various instances, the sensory augmentation platform **100** is also configured to receive inputs from a user via various devices or sensors discussed more fully herein. For example, the sensory augmentation platform **100** receives the inputs from the user responsive to the one or more smart rings **112** being in an active operational state, as discussed more fully herein.

[0031] In some examples, the one or more smart rings **112** are wearable apparatus configured to deliver haptic stimulation to a user's finger or fingers in order to communicate human-readable messages or other information through the haptic stimulation.

[0032] Each of the one or more smart rings **112** may include haptic stimulators **114** (e.g., vibrators, electrodes, and/or piezoelectric stimulators) configured to provide haptic stimulation to a user or users of the one or more smart rings **112** based on data corresponding to information to be communicated to the user as described in more detail below with reference to FIGS. 2A and 2B. Briefly, the one or more smart rings **112** cause the haptic stimulators **114** to provide haptic stimulation such that a user of the smart ring **112** feels the haptic stimulation at different locations or positions on the body (e.g., multiple locations of a finger), the locations corresponding to different information. For instance, haptic stimulation may be felt at different positions around a finger proximate to an interior annulus of a ring including the haptic stimulators **114**. By registering the different clock positions of the haptic stimulation, a user may register different messages.

[0033] Similarly, a user may feel different sensations arising from the haptic stimulation, the different character of the sensations corresponding to different information. For example, the haptic stimulation may provide different sensations of hardness, softness, pressure, vibration, and/or any other mechanical or non-mechanical sensory (e.g. electrical) input, as well as different combinations thereof. The different sensations may correspond to different information communicated to the user and may be combined with stimulation perceived to be at different locations, thereby communicating yet further varieties of information.

[0034] The one or more smart rings **112** may additionally include one or more network interfaces **127** to facilitate communication between the one or more smart rings **112** and other devices (such as the computing device **120** and the augmentation control server **124**). The network interfaces **127** may include a network interface **116** and/or a body area network interface **118**.

[0035] The network interfaces **116** may comprise an electronic modem (e.g. a radio frequency (RF) transceiver) or other communication device. For example, the network interfaces **116** may be or may include a first transceiver configured to connect the one or more smart rings **112** to a

network **122** (e.g., the Internet), may include a second transceiver configured to connect another smart ring **112** to other components of the sensory augmentation platform **100**, or may be or may include both the first transceiver and the second transceiver. In some examples, the network interfaces **116** may comprise a Bluetooth, Wi-Fi, WiMAX, 802.11x, ZigBee, cellular, or any other interface operative to communicatively couple the one or more smart rings **112** to the network **122**, and/or operative to communicatively couple the one or more smart rings **112** to one or more other components or devices in the sensory augmentation platform **100**.

[0036] The body area network interfaces **118** may provide Bluetooth or other connection that communicatively couples one or more aspects (e.g., components) of a smart ring **112** to one or more other aspects (e.g., components) of the smart ring **112**, communicatively couples each of the one or more smart rings **112** to one or more other smart rings **112**, and/or communicatively couples each of the one or more smart rings **112** to another device. The body area network interfaces **118** may provide a Bluetooth, Wi-Fi, WiMAX, 802.11x, ZigBee, cellular or other electrical, optical, sonic, or other operative connection between aspects of each of the one or more smart rings **112** and another smart ring (or another device). The body area network interfaces **118** may receive input from other aspects of the one or more smart rings **112** and may provide output to other aspects of the one or more smart rings **112**. One such aspect may include the network interface **116**. In some examples, the body area network interface **118** is omitted and the network interface **116** also performs the function of the body area network interface **118**.

[0037] During operation, the smart ring **112** may be in electronic communication with the network **122**. The smart ring **112** may also communicate with other smart rings **112** via the network **122**. For example, multiple individuals may wear smart rings **112**, which may be connected to the network **122**. The individuals may send and receive messages to/from the smart ring **112** of each other. Furthermore, a user may wear multiple smart rings **112**, such as the smart ring **112a** and **112b**. While in some embodiments, multiple smart rings **112** of a user may intercommunicate via a local network or virtual local area networks, such as via local Bluetooth connections to a user's smartphone, in further instances, multiple smart rings **112** of a user may both connect to the network **122** and may intercommunicate via the network **122**. Thus, the one or more smart rings **112** shown in FIG. 1 may be associated with different individuals and/or multiple smart rings **112** shown in FIG. 1 may be utilized by a same individual while the smart rings **112** are communicatively coupled to each other.

[0038] In various embodiments, during operation, the one or more smart rings **112** may be in electronic communication, via the network **122**, with an augmentation control server **124**. In various instances, the augmentation control server **124** provides data to one or more of the smart rings **112** corresponding to a message to communicate to a user via haptic stimulation. For example, the augmentation control server **124** may provide data **41a** or **41b** described below with reference to FIGS. 2A and 2B to one or more of the smart rings **112**. Moreover, in various instances, the augmentation control server **124** receives data, such as the transmission signals **44a** or **44b** of FIG. 2, regarding a user, such as via data input aspects of the one or more smart rings

112, for processing in connection with messages, and/or for communication to another user of a further smart ring 112. In an example, the augmentation control server 124 receives the data responsive to the one or more smart rings 112 being in an active operational state.

[0039] Each of the one or more smart rings 112 may be configured to communicate with the augmentation control server 124, and the augmentation control server 124 may be configured to permit or reject forwarding of a message to a different smart ring 112. In this manner, each of the one or more smart rings 112 may transmit information, receive information, and interact with other smart rings of the one or more smart rings 112, as desired. In various embodiments, a user may wear two of the one or more smart rings 112, or may wear a smart ring 112 with multiple worn components to stimulate multiple parts of a user's body. For instance, the one or more smart rings 112 may include a ring worn on each hand. The augmentation control server 124 may coordinate the provision of messages to the multiple smart rings 112 to be delivered in a time-coordinated manner to a user. In further instances, a local device of the user coordinates the provision of messages to the multiple smart rings 112 to be delivered in a time-coordinated manner to the user. For instance, one or more of the one or more smart rings 112 may effectuate such coordination.

[0040] The sensory augmentation platform 110 may include a visual device 26. The visual device 26 is configured to generate visual images for a user to view. For example, the visual device 26 may comprise a heads-up display on a pair of worn glasses. In further instances, the visual device 26 is a screen of the computing device 120 or a television. In various instances, text, photos, images, or other visual indications are provided to a user, by the one or more smart rings 112, such as to provide for mapping, augmented reality, or other visual services. Moreover, because a user may be unfamiliar with the meaning of different haptic stimuli, a representative image may accompany each stimulus. Also, in various training scenarios, the one or more smart rings 112 may, via the network interface 116 or the body area network interface 118, connect directly to a computing device, such as a computer or a smartphone. The computer or smartphone may run a training application, such as a game, scenario based training program, tutorial, or other application whereby a user is trained to interact with the one or more smart rings 112. For example, the one or more smart rings 112 may be configured to provide haptic stimuli to a finger representing a variety of different characters or words, while showing images of the relevant character or word on the visual device 26.

[0041] The computing device 120 may be a smartphone, a tablet, a wearable computer, a portable computer, such as a laptop, or any other computing device 120 as desired. In various embodiments, the computing device 120 is an integrated processor of the smart ring 112. In various embodiments, during operation the one or more smart rings 112 connect directly to the computing device 120. In various embodiments, the computing device 120 is operatively connected to one or more of the one or more smart rings 112 and the visual device 26 by a wireless technology. For instance, the visual device 26 may be a wearable pair of glasses with a computer therein.

[0042] In some examples, the computing device 120 provides processing power and memory for the one or more smart rings 112. For instance, the one or more smart rings

112 may be small, and/or power efficient, such that processing tasks are offloaded to a connected device, such as the computing device 120. In this manner, battery life of the one or more smart rings 112 may be improved and the form factor may be miniaturized for the one or more smart rings 112. In some examples, the one or more smart rings 112 ignoring human input while in the inactive operational state, and transmitting received human input, such as to the computing device 120, while in the active operational state further provides for increased battery life of the one or more smart rings 112 and reduced consumption of power by the computing device 120.

[0043] The processing power provided by computing device 120 may be used for a variety of purposes. For instance, data may be structured and rearranged to be suitable for provision to a smart ring 112. For example, a user may wish to read a text message from a friend via electrical stimulation signals provided to a finger by a smart ring 112 worn on that finger. The computing device 120 may process the text message character-by-character or word-by-word to generate symbols or other data corresponding to the message for delivery via haptic stimulation. The symbols or other data may be then provided by the computing device 120 to the smart ring 112 in sequence or to a plurality of the one or more smart rings 112 in parallel. Moreover, the computing device 120 may set the delivery rate of the stimulation (e.g., the timing described below with reference to FIG. 2), for instance, depending on a user's proficiency in interpreting electro-haptic messages. Further, and as mentioned above, in addition to text, any human perceptible information desired to be communicated may be translated to corresponding electrical stimuli. For instance, sounds, emoji and other representative characters, scents, graphics, and/or the like may be associated with a particular sensation or set of sensations communicable via electrical stimuli.

[0044] With reference to FIG. 2A, an example of a smart ring 212a is disclosed. The smart ring 212a of FIG. 2A may correspond to the smart ring 112a of FIG. 1. The smart ring 212a may be configured to be worn on a first part of the user's body. For example, the first smart ring 212a may be configured to be worn on a first index finger of a user.

[0045] The smart ring 212a may comprise a ring body 211a, one or more haptic stimulators 214a coupled to the ring body 211a, a communications interface 227a coupled to the ring body 211a, and a processor 218a coupled to the ring body 211a and the one or more haptic stimulators 214a and the communications interface 227a.

[0046] The ring body 211a is a wearable loop configured to be fitted around and/or worn by a user's finger. For example, the ring body 211a may be configured to be fitted around and/or worn by a user's index finger or other finger. In some examples, the ring body 211a is rigid and is formed of one or more rigid materials, such as a metal, rigid plastic, or a combination thereof, and may include a coating. In some examples, the ring body 211a is formed of titanium or ceramic zirconia. In some examples, the ring body 211a includes a diamond like carbon coating. In other examples, the ring body 211a is semi-rigid and is formed of a semi-rigid material. In some examples, the ring body 211a may form a complete loop. Alternatively, in other examples, the ring body 211a may form a partial loop. In various embodiments, the outside surface, the inside surface, or both may be substantially circular and smooth so as to increase user comfort when wearing the smart ring 212a.

[0047] The one or more haptic stimulators **214a** are any type of stimulator configured to (in response to one or more drive signals **48a** from the processor **218a**) provide haptic stimulation (e.g., vibration, electrical impulses, and/or pressure) to a user's body. For example, the one or more haptic stimulators **214a** may be or may include a vibrator, a piezo electronic configured to put pressure on the user's finger, an electrode configured to provide an electrical stimulation to the user's finger, or combinations thereof. In some examples, the one or more haptic stimulators **214a** include multiple haptic stimulators **214a**. In some examples in which the one or more haptic stimulators **214a** include multiple haptic stimulators **214a**, a number of the one or more haptic stimulators **214a** is greater than two. For example, the number of the one or more haptic stimulators **214a** may be within a range of 4-12 haptic stimulators.

[0048] In some examples in which the haptic stimulator **214a** includes multiple haptic stimulators **214a**, the processor **218a** may be configured to provide multiple of the one or more drive signals **48a**, and each of the multiple drive signals **48a** may correspond to a respective haptic stimulator **214a** of the multiple haptic stimulators **214a**. In these examples, each of the haptic stimulators **214a** is configured and coupled to the processor **218a** such that the haptic stimulator **214a** is configured to receive a corresponding drive signal of the multiple drive signals **48a**. As a result, in these examples, each of the haptic stimulators **214a** is independently excitable such that different combinations of the haptic stimulators **214a** can be excited.

[0049] The haptic stimulators **214a** are configured such that, during operation of the smart ring **212a**, the user may be able to feel haptic stimulation from the haptic stimulators **214a** at various locations on the user's body on which the smart ring **212a** is worn. The locations (or combinations thereof) of the haptic stimulation may correspond to different information. For instance, haptic stimulation may be felt at different positions around a finger proximate to an interior annulus of the smart ring **212a**. By registering the different positions of the haptic stimulation, a user may register different messages (e.g., content of the messages or types of the message).

[0050] Additionally or alternatively, the user may be able to register different messages by registering different feelings or sensations responsive to the haptic stimulation. For example, as described in more detail below, the haptic stimulators **214a** may include electrodes **244a** configured to apply stimulation waveforms or signals that are configured to induce sensations in one or more nerves of the user, and the sensations associated with the different nerves may register as different feelings or sensations with the user. By registering the different feelings or sensations responsive to the haptic stimulation, a user may register different messages.

[0051] Additionally or alternatively, the user may be able to register different messages by registering characteristics of the haptic stimulation, such as duration, pulsing, and/or intensity. For example, as described in more detail below, the haptic stimulators **214a** may include the vibration stimulators **242a**, and the smart ring **212a** may be configured to apply the haptic stimulation at different durations, pulses, and/or intensity. By registering the different characteristics of the haptic stimulation, a user may register different messages.

[0052] The haptic stimulators **214a** may include vibration stimulators **242a** (e.g., when the haptic stimulation includes vibration), electrodes **244a** (e.g., when the haptic stimulation includes electrical impulses), and/or piezoelectric stimulators **246a** (e.g., when the haptic stimulation includes pressure).

[0053] Each of the vibration stimulators **244a** is configured to, when actuated responsive to the drive signal from the processor **218a**, provide vibration, and may include an actuator (e.g., a motor) and a vibrating element that is configured to contact a skin of a user of the smart ring **212a** when the user wears the smart ring **212a**. Each of the piezoelectric stimulators **246a**, when driven responsive to the one or more drive signals **48a** from the processor **218a**, is configured to deform (e.g., expand or contract) and create pressure on a skin of the user when the user wears the smart ring **212a**.

[0054] Each of the electrodes **244a**, when driven responsive to the one or more drive signals **48a** from the processor **218a**, is configured to transfer an electrical impulse to a skin of the user of the smart ring **212a** when the user wears the smart ring **212a**. In various embodiments, multiple of the electrodes **244a** comprise a current source and one electrode of the electrodes **244a** comprises a current sink. In further embodiments, multiple of the electrodes **244a** comprise a current sink. In various embodiments, a single electrode of the electrodes **244a** comprises a current source and multiple of the electrodes **244a** comprise a current sink. In further embodiments, multiple of the electrodes **244a** comprise the current source. The electrodes **244a** may be configured to deliver electrical stimulation based on characteristics of the one or more drive signals **48a** from the processor **218a**. For example, the characteristics of the one or more drive signals **48a** may correspond to voltage, current, waveform (e.g., defining pulse length, wave shape such as square wave, triangle wave, sine wave, varying waveform, direct current (DC), duty cycle, etc.), and/or frequency, and the electrodes **244a** may be configured to deliver electrical stimulation having a stimulation waveform based on the voltage, current, waveform, and/or frequency of the one or more drive signals **48a**. Moreover, the electrodes **244a** may be configurable such that an electrode's **244a** role as anode or cathode may be changed and the polarity of the electrical signal(s) changed over time. Furthermore, the electrodes **244a** are configured such that different electrical signals may be delivered in parallel or in sequence, and combined. Yet furthermore, electrodes **244a** may be spaced, and different electrodes **244a** may be energized with different electrical signals (different stimulation waveforms). In addition, electrodes **244a** are configured such that they may be selectively driven so that spatial sensations (e.g., the perceived origin of the electrical stimulation) in addition to the character of the sensation itself may be changed to communicate different messages. The electrodes **244a** may be configured such that an electrical signal may be passed between the electrodes **244a**. Moreover, multiple electrical signals may be passed between or among the electrodes **244a**.

[0055] The electrodes **244a**, when driven by the one or more drive signals **48a**, are configured to deliver electrical stimulation to one or more nerves or types of nerves that can be electrically stimulated to induce sensations, such as sensations of mechanical or electrical stimuli. The human body contains a variety of different nerves. Many such nerves are capable of detecting mechanical or electrical

stimulus. Moreover, it has been determined that many such nerves can be electrically stimulated to induce sensations, including sensations of mechanical stimuli. The various nerves may be classified. For example, different stimuli may be perceived by different nerves. By stimulating the different nerves, different sensations may be evoked. Moreover, different nerves perceive stimuli in different spatial relation, thus different nerves may be stimulated as well, to communicate differing messages.

[0056] For instance, FIG. 3 illustrates four example types of nerves (a Meissner corpuscle **12**, a Merkel cell **14**, a Ruffini ending **16**, and a Pacinian corpuscle **18**) to which one or more of the electrodes **244a** may be configured to deliver electrical stimulation during operation of the smart ring **212a**. With reference to FIG. 3, the human skin may have a stratum corneum **4** overlaying the epidermis **6**. Various nerves are present in the epidermis **6**. Moreover, the epidermis **6** may overlay the dermis **8**. Various nerves may also be located in the dermis **8**. For context, two sweat ducts **10** are shown as well. In various instances, the Merkel cell **14** is activated by static pressure. In various instances, the Meissner corpuscle **12** is activated by time-varying vibrations, as is the Pacinian corpuscle **18**. In various instances, one may appreciate that the Ruffini ending **16** and the Pacinian corpuscle **18** may have different resonant frequencies. For instance, the Ruffini ending **16** may resonate at 120 hertz (Hz) and the Pacinian corpuscle **18** at 250 Hz. Moreover, myelinated nerves may comprise insulated nerves covered by a myelin sheath with gaps spaced at Nodes of Ranvier.

[0057] The different properties of the different nerve aspects may be exploited by different types of electrical stimulation, such that stimulation having both time variant, mode variant (e.g., the type of nerves activated), location variant, intensity variant, and sequence variant, and/or other path variant characteristics can communicate large character sets and immense amounts of human-readable information. Thus, by stimulating the different nerves with different electrical signals having different waveforms, amplitude, current, and the like, it is possible to induce different sensations. For example, the Merkel cell **14** receptors may encode pressure through firing frequency. Thus, by stimulating a Merkel cell **14** receptor with a variety of different electrical impulses of different frequencies, it may also be possible to induce sensations of touch of a hard object and/or touch of a soft object. For further example, a Meissner corpuscle **12** may be stimulated to elicit a stable vibratory sensation. Moreover, stimulation of the Pacinian corpuscle **18** may facilitate the generation of location-shifted stimuli, meaning the perceived stimulus is perceived to be at a different location that the electrode applying the electrical stimuli. Thus, any and all perceptible aspects of haptic stimulation may be evoked, in addition to unique, non-haptic stimulation.

[0058] Returning to FIG. 2A, the electrodes **244a** may be configured such that different combinations of electrodes **244a** may be selectively excited in different patterns and with signals of different voltage, current, waveform, frequency, and/or the like so that different sensations are perceived by the user and thus can be translated by the user's nervous system into different messages. For example, a long pulse followed by a short pulse followed by a long pulse, followed again by a short pulse may correspond to the character "C," whereas a short pulse followed by a long pulse may correspond to the character "A." More complex

symbolic representations are possible due to the many possible combinations of pulse length, possibilities of which electrode or electrodes of the electrodes **244a** is stimulated, and possibilities of which combinations of electrodes **244a** are stimulated, the voltage, current waveform, and/or frequency of each stimulation, both in sequence and in parallel, and/or the like.

[0059] The communications interface **227a** may correspond to the network interface **127** described above with reference to FIG. 1. In some examples, the communications interface **227a** includes the network interface **116** of FIG. 1, the body area network interface **118** of FIG. 1, or any combination thereof.

[0060] The communications interface **227a** is configured to receive data **41a** corresponding to information to be communicated to a user. The information to be communicated to the user may include text, images (e.g., screen content of the computing device **120** of FIG. 1), graphics, memes, phonemes, calculations, sounds, emoji, alerts, and/or any other information able to be received through other human senses. As an example, the communications interface **227a** may be configured to receive the data **41a** from the computing device **120** of FIG. 1, the augmentation control server **124** of FIG. 1, or both.

[0061] In some examples, the data **41a** may include or indicate the information to be communicated to the user. For example, the information to be communicated to the user may include text, and the data **41a** may include or indicate the text.

[0062] Alternatively or additionally, the data **41a** may be based on the information to be communicated to the user but may not include or indicate the information to be communicated to the user. In these examples, the data **41a** may indicate, identify, and/or instruct when and/or how the smart ring **212a** should drive the haptic stimulators **214a** in order to communicate the information to be communicated to the user. For example, the computing device **120** of FIG. 1 and/or the augmentation control server **124** of FIG. 1 may determine, based on the information to be communicated to the user, information corresponding to one or more drive signals **48a** that the smart ring **212a** should provide one or more of the haptic stimulators **214a**, timing at which the smart ring **212a** should apply the one or more drive signals **48a**, particular stimulators of the haptic stimulators **214a** that the smart ring **212a** should drive at any given time, intensity of the one or more drive signals **48a**, duration of the one or more drive signals **48a**, and/or the sequence at which the particular stimulators should be driven, in order to communicate the information to be communicated to the user using the haptic stimulators **214a**. In these examples, the data **41a** may include data that indicates, that identifies, and/or that includes instructions to cause the smart ring **212a** to drive one or more of the haptic stimulators **214a** according to, the information corresponding to one or more drive signals **48a**, the timing, the particular haptic stimulators, the intensity, the duration, and/or the sequence, in order to communicate the information to be communicated to the user.

[0063] For example, the haptic stimulators **214a** may include the vibration stimulators **242a**, and the data **41a** may indicate signal characteristics, timing (e.g., intervals between vibrations corresponding to different elements of the information), particular vibration stimulators **243a** to activate, voltage, duration, and/or the sequence at which to

activate the particular vibration stimulators **243a** in order to communicate the information to be communicated to the user. For example, the data **41a** may indicate first particular vibration stimulators **51a** to drive at a first time using the one or more drive signals **48a**, and may indicate second particular vibration stimulators **53a** to drive at a second time using the drive signals **48a**, and the second particular vibration stimulators **53a** may be different than the first particular vibration stimulators **51a**.

[0064] As another example, the haptic stimulators **214a** may include the electrodes **244a**, and the data **41a** may indicate signal characteristics (e.g., pulse duration, voltage), particular electrodes **245a** to activate at any given time, timing, and the sequence at which to activate the particular electrodes **245a** in order to communicate the information to be communicated to the user. For example, the data **41a** may indicate first particular electrodes **55a** to drive at a first time using the one or more drive signals **48a**, and may indicate second particular electrodes **57a** to drive at a second time using the one or more drive signals **48a**, and the second particular electrodes **57a** may be different than the first particular electrodes **55a**.

[0065] As another example, the haptic stimulators **214a** may include the piezoelectric stimulators **246a**, and the data **41a** may indicate signal characteristics (e.g., pulse duration, voltage), particular piezoelectric stimulators **247a** to activate at any given time, and the sequence at which to activate the particular piezoelectric stimulators **247a** in order to communicate the information to be communicated to the user. For example, the data **41a** may indicate first particular piezoelectric stimulators **59a** to drive at a first time using the one or more drive signals **48a**, and may indicate second particular piezoelectric stimulators **61a** to drive at a second time using the one or more drive signals **48a**, and the second particular piezoelectric stimulators **61a** may be different than the first particular piezoelectric stimulators **59a**.

[0066] The processor **218a** is configured to receive the data **41a** and to generate and provide the one or more drive signals **48a** (e.g., control signals, drive instructions) to the one or more haptic stimulators **214a** based on the data **41a**. The processor **218a** may be an embedded computing device, a processor and memory, or any other programmable electronic device or circuit configured to receive the data **41a** and to send the drive signal **48a** to the one or more haptic stimulators **214a**. The drive signal **48a** is configured to drive or activate the haptic stimulator **214a** to communicate the information to be communicated to the user. In some examples, as explained above, the haptic stimulator **214a** includes multiple haptic stimulators **214a**. In these examples, the processor **218a** may be configured to provide multiple drive signals **48a**, and each of the multiple drive signals **48a** may correspond to a respective haptic stimulator **214a** of the multiple haptic stimulators **214a**. In these examples, each of the haptic stimulators **214a** is configured and coupled to the processor **218a** such that the haptic stimulator **214a** is configured to receive a corresponding drive signal **48a** of the multiple drive signals **48a**. As a result, in these examples, each of the haptic stimulators **214a** is independently excitable such that different combinations of the haptic stimulators **214a** can be excited.

[0067] In some examples, as explained above, the data **41a** may include or may indicate the information to be communicated to a user. In these examples, the processor **218a** may be configured to translate the data **41a** to determine which of

the haptic stimulators **214a** to drive and when and/or how the smart ring **212a** should drive the haptic stimulators **214a** in order to communicate the information to be communicated to the user. For example, the processor **218a** may determine, based on the information to be communicated to the user included or indicated in the data **41a**, characteristics of the drive signals **48a** that the smart ring **212a** should provide one or more of the haptic stimulators **214a**, timing at which the smart ring **212a** should apply the drive signals **48a**, duration of the drive signals **48a**, intensity of the drive signals **48a**, and/or particular haptic stimulators **214a** that the smart ring **212a** should drive at any given time, in order to communicate the information to be communicated to the user. In these examples, subsequent to determining the characteristics of the drive signals **48a**, the timing, the duration, the intensity, and/or the particular haptic stimulators **214a**, the processor **218a** generates the drive signals **48a** to drive the particular haptic stimulators **214a** according to the determined characteristics, timing, duration, and intensity.

[0068] In some examples in which the data **41a** includes or indicates the information to be communicated to the user, the information to be communicated to the user includes text. In some of these examples, the processor **218a** is configured to process the data **41a** to generate the drive signals **48a** to drive different haptic stimulators **214a** for different characters of the text. For example, each letter of an alphabet (e.g., the English alphabet) may correspond to a different pattern of activated haptic stimulators **214a**, and the processor **218a** may be configured to generate the drive signals **48a** based on the letters in the text and the patterns corresponding to the letters in the text to drive particular haptic stimulators of the haptic stimulators **214a** in a sequence corresponding to the sequence of the letters in the text and according to the patterns corresponding to the letters in the text. In some examples, the patterns of haptic stimulators corresponding to letters in an alphabet may be based on or correspond to vertices or other index points on a represented character.

[0069] In an example, the drive signals **48a** are configured to drive one or more first haptic stimulators of the one or more haptic stimulators **214a** to communicate a first character of the text. In this example, the drive signals **48a** are also configured to drive one or more second haptic stimulators of the one or more haptic stimulators **214a** to communicate a second character of the text. In this example, the one or more second haptic stimulators include at least one haptic stimulator that the one or more first haptic stimulators do not include.

[0070] To illustrate, the information to be communicated to a user may include the word “Apple,” and the data **41a** may include the word “Apple” or may otherwise indicate that the word “Apple” is to be communicated to the user. In this example, the processor **218a** may be configured to determine characteristics of the drive signals **48a**, the timing, and/or the particular haptic stimulators **214a** as described below with reference to FIGS. 4A-4F based on receiving the word “Apple” included in or otherwise indicated by the data **41a**. In this example, the processor **218a** then generates the drive signals **48a** according to the characteristics, the timing, and/or the particular haptic stimulators **214a** determined according to the data **41a** indicating or identifying the word “Apple” and described below with reference to FIGS. 4A-6F, and provides the drive signals **48a** to the particular haptic stimulators **214a** described below

with reference to FIGS. 4A-6F according to the timing described below with reference to FIGS. 4A-6F.

[0071] Returning to FIG. 2A, the data 41a may indicate, identify, and/or instruct which of the haptic stimulators 214a to drive and when and/or how the smart ring 212a should drive the haptic stimulators 214a in order to communicate the information to be communicated to the user. In these examples, the data 41a may include data that indicates, that identifies, and/or that includes instructions to cause the sensory augmentation system 212a to drive one or more of the haptic stimulators 214a according to, characteristics of the drive signals 48a, timing, duration, intensity, and/or particular haptic stimulators 214a, indicated or identified by the data 41a in order to communicate the information to be communicated to the user. In these examples, the processor 218a processes the data 41a to obtain the characteristics, the timing, the duration, the intensity, and/or the particular haptic stimulators 214a indicated, identified, and/or instructed by the data 41a.

[0072] To illustrate, the information to be communicated to a user may include the word “Apple,” and the data 41a may identify or indicate characteristics of the drive signals 48a, timing, duration, intensity, and particular haptic stimulators 214a described below with reference to FIGS. 4A-4F. In this example, the processor 218a may be configured to process the data 41a to obtain the characteristics, the timing, the duration, the intensity, and/or the particular haptic stimulators 214a indicated by the data 41a and described below with reference to FIGS. 4A-4F, and may generate the drive signals 48a to drive the particular haptic stimulators 214a described below with reference to FIGS. 4A-4F using the drive signals 48a according to the timing described below with reference to FIGS. 4A-4F.

[0073] In some examples, because the haptic stimulators 214a typically will require a higher voltage electrical waveform than many microprocessors generate, in various embodiments, the processor 218a includes and/or uses an H-bridge, or a buck/boost converter, or one or more capacitors for discharging, or one or more transformers, coils, and/or the like to generate the drive signals 48a.

[0074] The smart ring 212a may include a power source 280a. The power source 280a comprises a source of electrical energy. The power source 280a may be a battery, or a super capacitor. The power source 280a may include a charger, such as to harvest RF energy, body heat, motion of the user, and/or the like. The power source 280a may include an inductive charger, such as to facilitate recharging of a battery or super capacitor by exposure to an electromagnetic field. The power source 280a is coupled to the processor 218a, and may be configured to enable the processor 218a to drive the one or more haptic stimulators 214a with sufficient energy to activate the one or more haptic stimulators 214a.

[0075] In some examples, the smart ring 212a may additionally include an effector 281a. The effector 281a may comprise a visual indicator, such as a light source, or an electronically-controlled display or pattern provided by a liquid crystal display (LCD), electronic ink and/or any type of human perceptible indicator as desired. The effector 281a may further comprise non-human readable mechanisms, such as a near field communications (NFC) device configured to effect financial transactions and/or any other function as desired. In some embodiments, the effector 281a may be combined with the touch sensor 52a.

[0076] In some examples, the smart ring 212a additionally or alternatively includes one or more user sensors 50a coupled to the ring body 211a and configured to detect input 62a from a user of the smart ring 212a. The one or more user sensors 50a are configured to convert the input 62a into an input signal 42a and provide the input signal 42a to the processor 218a. The one or more user sensors 50a comprise a touch sensor 52a, one or more position sensors 54a, a microphone 56a, one or more biometric sensors 58a, an optical sensor 59a, a magnetic sensor 61a, or any combination thereof. The one or more user sensors 50a may be coupled to an exterior and/or interior side of the ring body 211a and/or may be embedded within the ring body 211a. For example, the touch sensor 52a may include one or more touch sensors 52a disposed on an exterior annular surface of the ring body 211a. As another example, the position sensor 54a may include one or more position sensors 54a coupled to or embedded within the ring body 211a. As another example, the microphone 56a may be coupled to an exterior annular surface of the ring body 211a. As another example, the biometric sensors 58a may include one or more biometric sensors disposed on an interior surface of the ring body 211a.

[0077] The touch sensor 52a may be any type of sensor configured to detect touch input from a user. For example, the touch sensor 52a may be a capacitive touch sensor. In examples in which the one or more user sensors 50a include the touch sensor 52a, the input 62a from the user of the smart ring 212a includes touch input. In these examples, the input signal 42a includes a location, number, and/or duration of the touch inputs. In some examples, the touch input 62a comprises one of a single tap, a double tap, or a deep tap. The touch sensor 52a may be configured such that different locations on the touch sensor 52a correspond to respective information input, such as human readable message elements or control instructions. In some examples, a user may stimulate the touch sensor 52a such as by forming a hand having the smart ring 212a into a fist, and using a finger of another hand to touch the external annular surface of the smart ring 212a.

[0078] For example, the touch sensor 52a may be configured such that different locations on the touch sensor 52a correspond to respective human readable message elements, and the user may provide touch input at locations of the touch sensor 52a that correspond to the human readable message elements the user desires to include in a message (e.g., a short message service (SMS) message). For example, the user may wish to communicate a text message formed of text characters, and may provide touch input on the touch sensor 52a at locations corresponding to the text characters. The smart ring 212a may thus be configured to receive, via touch input, human readable message elements of a message the user desires to communicate.

[0079] As another example, the touch sensor 52a may be configured such that different locations on the touch sensor 52a correspond to respective control instructions for controlling the smart ring 212a or another device, and the user may provide touch input at locations of the touch sensor 52a that correspond to the control action the user desires to effect. To illustrate, the user may wish to disable communication using the ring, and may provide touch input on the touch sensor 52a at a location of the touch sensor 52a that corresponds to disabling communication on the smart ring 212a. As another example, the user may wish to disable the

haptic stimulators on the smart ring **212a**, and the user may provide touch input on the touch sensor **52a** at a location of the touch sensor **52a** that corresponds to disabling the haptic stimulators **214a**. As another example, the smart ring **212a** may serve as a remote control to change the channel of a television. In this example, different locations of the touch sensor **52a** may correspond to respective control instructions for controlling the television, and the user may provide touch input at a location on the touch sensor **52a** that corresponds to changing the channel on the television. The smart ring **212a** may thus be configured to receive, via touch input, control instructions for controlling the smart ring **212a** or another device.

[0080] In some examples, the touch sensor **52a** is further configured to identify a fingerprint of the user and transmit user fingerprint data to the processor **218a**. In some examples, the user fingerprint data is used to power on, power off, or unlock the smart ring **212a**.

[0081] In some examples, the touch sensor **52a** additionally or alternatively includes a capacitive touch screen that can display messages to the user.

[0082] The position sensor **54a** may be any type of sensor configured to obtain position data. For example, the position sensor **54a** may be or may include one or more accelerometers, a gyroscope, or a combination thereof. The position sensor **54a** may be configured to detect motion in six degrees of freedom. For example, the position sensors **54a** may include accelerometers that can calculate movement of the ring in three dimensions of a Cartesian coordinate system, as well as rotation about any of the Cartesian coordinate axes (e.g. roll, pitch and yaw). The position sensors may also include global positioning system (GPS), wireless (e.g. WiFi or cellular) triangulation, or any other means to determine position. In examples in which the one or more user sensors **50a** include the position sensor **54a**, the input **62a** from the user of the smart ring **212a** includes motion or position information (e.g., position data) of the smart ring **212a** (and correspondingly motion or position information a user of the smart ring **212a**) from the one or more accelerometers and/or the gyroscope. In these examples, the input signal **42a** includes or indicates motion corresponding to the motion or position information. In these examples, the input signal **42a** includes position changes in three dimension of a Cartesian coordinate system.

[0083] In some examples, the input **62a** comprises motion information of the user corresponding to human readable message elements or to control instructions regarding the smart ring **212a** or another device. For example, different motions may correspond to respective human readable message elements, and the position sensor **54a** may be configured to detect the different motions. To illustrate, different hand motions corresponding to, for example, sign language, may correspond to respective words, and the position sensor **54a** may be configured to detect motion corresponding to the different hand motions. The smart ring **212a** may thus be configured to receive, via motion detection, human readable message elements of a message the user desires to communicate.

[0084] As another example, different hand motions may correspond to respective control instructions. To illustrate, a shake or vibration hand motion may correspond to disregarding an incoming message, and the position sensor may be configured to detect the shake hand motion. The smart

ring **212a** may thus be configured to receive, via motion detection, control instructions to control the smart ring **212a** or another device.

[0085] The microphone **56a** may be configured to detect sound. In examples in which the one or more user sensors **50a** comprise the microphone **56a**, the input **62a** from the user of the smart ring **212a** includes voice input. In these examples, the input signal **42a** includes or indicates voice data corresponding to the voice input. The voice input may correspond to human readable message elements that the user desires to include in a message. Alternatively or additionally, the voice input may correspond to control instructions for controlling the smart ring **212a** or another device.

[0086] The biometric sensors **58a** may be any sensor configured to detect biometric data of a user, such as temperature, heart rate, blood pressure, oxygen level, perspiration level, breathing rate, steps taken, or any combination thereof. In examples in which the one or more user sensors **50a** include the biometric sensors **58a**, the input **62a** comprises biometric data of the user. In these examples, the input signal **42a** includes or indicates the biometric data.

[0087] The optical sensor **59a** is oriented outward from the ring such that the optical sensor **59a** is oriented towards an end of the finger when the smart ring **212a** is worn by the user. The optical sensor **59a** may include a light source, a measuring device, a camera, and/or a sensor (e.g., a photo-detector configured to receive a pre-determined wavelength range (e.g. the visual spectrum)). The sensor of the optical sensor **59a** is configured to sense light received at the optical sensor **59a** (e.g., light that was transmitted by the light source and reflected by the item or light that was transmitted by a light source on the item) and measure the light. The optical sensor **59a** is configured to provide information corresponding to the measured light to the processor **218a** as the input signal **42a**.

[0088] The magnetic sensor **61a** is configured to detect the presence of other nearby smart rings.

[0089] As described above, in some examples the smart ring **212a** is configured to ignore, or not to form, the input signal **42a**. For example, the smart ring **212a** may include active and inactive operational states. In some implementations, a default state of the smart ring **212a** may be the inactive operational state. While operating in the inactive operational state, in one example, the smart ring **212a** may form the input signal **42a** but not provide the input signal **42a** to the processor **218a**. In another example, while operating in the inactive operational state the smart ring **212a** may disable any one or more of the touch sensor **52a**, the motion/position sensor **54a**, the microphone **56a**, the biometric sensor **58a**, the optical sensor **59a**, or the magnetic sensor **61a**.

[0090] Not forming the input signal **42a**, not providing the input signal **42a** to the processor **218a**, or disabling one or more user sensors **50a** may preserve power of the smart ring **212a**. For example, during some times of operation a user may wish for data obtained via one or more user sensors **50a** to be formed into the input signal **42a** and provided to the processor **218a**. In other examples, the user may not wish to have the data obtained via one or more user sensors **50a** to be formed into the input signal **42a** and provided to the processor **218a**. The smart ring **212a** being configured with both inactive and active operational states provides the user with control over the circumstances in which the input signal **42a** is formed. For example, not all data captured by the one

or more user sensors **50a** may be intended by the user to be input into a system in which the smart ring **212a** is implemented. By not obtaining data via at least some of the one or more user sensors **50a**, and/or forming and providing the input signal **42a** during periods of time in which the user wishes the input signal **42a** not to be provided, reduces power consumption of the smart ring **212a** in capturing or processing data that is not intended to be used as input to the system of the smart ring **212a**. In yet other examples, data may be captured by one or more user sensors **50a** and provided to the processor **218a**, but may not be transmitted to a device other than the smart ring **212a** while the smart ring **212a** is in the inactive operational state.

[0091] Responsive to the smart ring **212a** being placed in the active operational state, the smart ring **212a** is configured to capture data via one or more user sensors **50a**, form the input signal **42a**, provide the input signal **42a** to the processor **218a**, and in some examples, provide a signal based on the captured data to a device other than the smart ring **212a**. To place the smart ring **212a** in the active operational state, a user may provide a programmed input to the smart ring **212a**, such as via at least one of the one or more user sensors **50a**. For example, the user may provide a single discreet input or multiple inputs so as to form a pattern. The input(s) can include touch input(s), audio input(s), biometric input(s), or the like. Responsive to receipt of an input that causes the smart ring **212a** to be in the active operational state, the smart ring **212a** captures data via the one or more user sensors **50a**, such as magnetic data and/or motion/position data. Responsive to removal of the input causing the smart ring **212a** to be in the active operational state, or receipt of a second input configured to place the smart ring **212a** in the inactive operational state, the smart ring **212a** may operate in the inactive operational state, as described elsewhere herein. In other examples, the active operational state may extend for a programmed period of time, such as about 3 seconds, about 5 seconds, about 7 seconds, about 10 seconds, etc. Upon expiration of the programmed period of time, the smart ring **212a** may return to the inactive operational state.

[0092] In examples in which the smart ring **212a** includes the one or more user sensors **50a**, and subject to active/inactive operational state programming of the smart ring **212a**, the processor **218a** is additionally or alternatively configured to receive the input signal **42a** from the one or more user sensors **50a** and generate one or more transmission signals **44a** based on the input signal **42a**. The processor **218a** may be configured to generate the one or more transmission signals **44a** based on the input signal **42a** by packaging the input signal **42a** for transmission to a second device, such as the computing device **120** of FIG. 1 or the augmentation control server **124** of FIG. 1. Additionally or alternatively, in examples in which the one or more user sensors **50a** include the optical sensor **59a**, the processor **218a** is configured to generate the one or more transmission signals **44a** to include the information corresponding to the measured light. The processor **218a** is configured to provide the one or more transmission signals **44a** to the communications interface **227a**. The communications interface **227a** is configured to receive the one or more transmission signals **44a** from the processor **218a**, and to transmit the one or more transmission signals **44a** to the second device. The communications interface **227a** may use the human body interface

116 of FIG. 1, the network interface **118** of FIG. 1, or both, to transmit the one or more transmission signals **44a** to the second device.

[0093] The ring **212a** may additionally include one or more external ports (e.g., one or more micro-USB ports, etc.) for charging and wired communication.

[0094] In some examples, the smart ring **212a** may additionally or alternatively be configured to enable access to a second device. In these examples, the communications interface **227a** is additionally or alternatively configured to receive data **72a** indicating a near field frequency associated with access permission for accessing a second device. The second device may be any device that is accessed (e.g., unlocked, started) by or using (or that includes a function accessed by or using) a key (e.g., a radio frequency (RFID) key). For example, the second device may include an automated door lock. As another example, the second device may include a vehicle or a vehicle's engine. The data **72a** may be received from the key or from another device.

[0095] In examples in which the smart ring **212a** is configured to enable access to a second device, the processor **218a** is configured to obtain the near field frequency based on the data **72a** by processing the data **72a**. The smart ring **212a** may further be configured to enable accessing the second device by transmitting, using the communications interface **227a**, a signal **63a** corresponding to the near field frequency. Transmitting the signal **63a** corresponding to the near field frequency when the smart ring **212a** is within a threshold distance of the second device enables accessing the second device. For example, transmitting the signal corresponding to the near field frequency when the smart ring **212a** is within a threshold distance from a vehicle may enable a user to start a vehicle by pushing a start button on the vehicle without the vehicle key being within the threshold distance from the vehicle.

[0096] In some examples, the smart ring **212a** further includes a memory **283a** coupled to the processor **218a** and including an access identifier (ID) **285a**. In these examples, the communications interface comprises a near field communications (NFC) interface that is configured to communicate the access ID **285a** to a security reader external to the smart ring **212a**.

[0097] FIG. 2B illustrates an example of a smart ring **212b**. The smart ring **212b** may correspond to any smart ring of the one or more smart rings **112** of FIG. 1. For example, the smart ring **212b** may correspond to the smart ring **112b** of FIG. 1. The smart ring **212b** may be configured to be worn on a second part of the user's body (e.g., a different finger than the finger on which the smart ring **212a** of FIG. 2A is worn). The smart ring **212b** may comprise a ring body **211b**, haptic stimulators **214b** coupled to the ring body **211b**, a communications interface **227b** coupled to the ring body **211b**, a processor **218b** coupled to the ring body **211b**, haptic stimulators **214b** coupled to the processor **218b**, a memory **283b** coupled to the processor **218b**, one or more user sensors **50b** coupled to the processor **218b**, and a communications interface **227b** coupled to the processor **218b**.

[0098] The ring body **211b** may be configured as described above with reference to the ring body **211a**, the haptic stimulators **214b** may be configured as described above with reference to the haptic stimulators **214a**, the communications interface **227b** may be configured to perform any of the functions described above with reference to the communications interface **227a**, and the processor **218b** may be

configured to perform any of the functions described above with reference to the processor **218b**.

[0099] The smart ring **212b** may be configured to receive data **41b**. The data **41b** may correspond to the data **41a**. The processor **218a** may be configured to process the data **41b** to generate one or more drive signals **48b** that correspond to the one or more drive signals **48a** described above with reference to the smart ring **212a**. The processor **218a** may be configured to provide the one or more drive signals **48b** to drive the one or more haptic stimulators **214b** as described above with reference to the processor **218a** and the one or more haptic stimulators **214a** of the smart ring **212a**.

[0100] In some examples, the data **41b** additionally or alternatively includes data that indicates that the smart ring **212a** is going to communicate (e.g., imminently communicate) the information described above with reference to the smart ring **212a** and/or that indicates the type of the information (e.g., text message). In these examples, the processor **218b** is configured to process the data **41b** to generate the drive signals **48b**, and may provide the drive signals **48b** to the haptic stimulators **214b** to communicate that the smart ring **212a** is going to communicate (e.g., imminently communicate) the information described above with reference to the smart ring **212a** and/or that indicates the type of the information (e.g., text message). In this manner, the smart ring **212a** may be configured to communicate content of a message, and the smart ring **212b** may be configured to communicate that the smart ring **212a** is going to receive content of the message and to communicate the type of the message.

[0101] The one or more user sensors **50b** that correspond to or function as described above with reference to, the one or more user sensors **50a**. For example, the smart ring **212b** may additionally include a touch sensor **52b**, a position sensor **54b**, a microphone **56b**, a biometric sensor **58b**, an optical sensor **59b**, and/or a magnetic sensor **61b**, that respectively correspond to the touch sensor **52a**, the position sensor **54a**, the microphone **56a**, the biometric sensor **58a**, the optical sensor **59a**, and/or the magnetic sensor **61a** described above with reference to the smart ring **212a**.

[0102] The smart ring **212b** additionally or alternatively includes an effector **281b** coupled to the processor **218b**, and a power source **280b** coupled to the processor **218b**. The effector **281b** may be configured as described above with reference to the effector **281a** of FIG. 2A, and the power source **280b** may be configured as described above with reference to the power source **280a**. The smart ring **212** may be configured to receive input **62b** as described above with reference to the input **62a** of FIG. 2A. The one or more user sensors **50b** may be configured to generate an input signal **42b** as described above with reference to the input signal **42a** of FIG. 2A. The smart ring **212b** may be configured to receive the signal **63b** as described above with reference to the signal **63a** of FIG. 2A. The smart ring **212b** may additionally or alternatively be configured to receive data **72b** as described above with reference to the data **72a** of FIG. 2A.

[0103] With reference to FIGS. 4A-4F, FIG. 4A illustrates example rings **412a** and **412b** including examples of haptic stimulators **414a** and **414b**, and FIGS. 4B-4F illustrate an example sequence of the haptic stimulators **414a** that may be activated by the ring **412a**. The rings **412a** and **412b** of FIG. 4A may correspond to the one or more smart rings **112** of FIG. 1 and/or to the smart rings **212a** and **212b** of FIGS. 2A

and 2B. The haptic stimulators **414a** and **414b** of FIG. 4A may correspond to the haptic stimulators **214a** or **214b** illustrated in FIGS. 2A and 2B. The haptic stimulators **414a** and **414b** are evenly spaced around the interior annulus of the ring bodies of the rings **412a** and **412b**. In some examples, two rings may be used so that two rings **412a** and **412b** are controlled in concert. For instance, one ring **412a** may communicate to the user the nature and character of the content presently or imminently communicated by the other ring **412b**. For instance, one ring **412b** may communicate to the user that an incoming text message has been received, while the other ring **412a** may communicate the content of the text message simultaneously or imminently thereafter.

[0104] The haptic stimulators **414a** and **414b** may comprise twelve haptic stimulators. In further instances, any number of haptic stimulators **414a** and **414b** may be selected. The haptic stimulators **414a** and **414b** may correspond to different locations around the ring body. For ease of viewing, the reference numbers **414a** and **414b** are only shown in connection with some of the twelve haptic stimulators **414a** and **414b** illustrated in FIGS. 4A-4F.

[0105] In various instances, an English word is desired to be communicated to a user. For instance, the word “APPLE” is desired to be communicated. While the haptic stimulators **414a** may communicate symbols, phonemes, or any other communicative device representable by haptic stimulation, in this non-limiting example, individual characters are communicated in sequence to comprise a word. For instance, FIG. 4B shows three haptic stimulators being energized. The haptic stimulators correspond to vertices or other index points on a represented character. For instance, for the letter A, haptic stimulators **414a** at positions 0, 7, and 5 may be energized. The number associated with the haptic stimulator corresponds to its clock position in this example, so the haptic stimulators **414a** at the 12 o'clock, 5 o'clock, and 7 o'clock positions are activated, corresponding to the three ends of the letter “A”. At a second moment in time, a different set of haptic stimulators **414a** may be activated to correspond to the second letter in the word, “APPLE.”

[0106] With reference to FIGS. 4A and 4C, haptic stimulators **414a** at positions 5 and 0 remain activated, and the haptic stimulator **414a** at position 7 is momentarily pulsed off then back on, and the haptic stimulator at position 11 is activated. In this manner, one may appreciate that the next letter, “P” is indicated, while the haptic stimulators **414a** associated with the first letter, “A,” in some embodiments, remain activated to facilitate remembering by the user of the earlier character.

[0107] Turning now to FIGS. 4A and 4D, one may note that a second letter “P” is desired to be indicated. However, in order to indicate two identical characters in a row, a further temporal aspect is implemented. Specifically, the haptic stimulators **414a** corresponding to the letter P are pulsed off and then on again to represent a second sequential instance of the character. Haptic stimulators **414a** previously associated with the letter “A” and not with the letter “P” may be left remaining active and not pulsed.

[0108] Subsequently, and with reference to FIGS. 4A and 4E, the next character, “L,” is communicated. Haptic stimulators **414a** at positions 11 and 7 are pulsed off and back on, as these haptic stimulators **414a** were also associated with the letter “P” and the haptic stimulator **414a** at position 5 is activated. The haptic stimulator **414a** at position 0 has remained activated and is unchanged.

[0109] Finally, shifting focus to FIGS. 4A and 4F, the final letter, “E,” is communicated. Haptic stimulators 414a at positions 3, 1, 11, 7, and 5 are all associated with the letter “E.” Haptic stimulators 414a at positions 11, 7, and 5 are pulsed off and back on, haptic stimulators 414a at positions 3 and 1 are activated, and the haptic stimulator 414 at position 0 remains activated and is unchanged.

[0110] In various embodiments, each letter, character, phoneme, or other symbol is associated with an equal-duration period of time (a “tick”). Each tick may be of a length sufficient to permit the user to identify the character, phoneme, or other symbol being communicated. Thus, the tick length may be shortened as a user gains proficiency. In various embodiments, following the final letter of a word or other collection of symbolically represented information, all haptic stimulators 414 are deactivated for a single tick, signaling the end of the word or other collection of symbolically-represented information. With reference to FIG. 4F, following the elapse of one tick for the final letter “E,” each haptic stimulator 414a may be deactivated for one tick, signaling the end of the word, prior to beginning presentation of subsequent user readable stimuli.

[0111] FIG. 5 illustrates an example of a smart ring 512. The smart ring 512 may correspond to the smart ring 112 of FIG. 1, the smart ring 212a of FIG. 2A, the smart ring 212b of FIG. 2B, or the smart ring 412 of FIG. 4A. The smart ring 512 includes haptic stimulators 514. The haptic stimulators 514 may correspond to the haptic stimulators 114 of FIG. 1 or to the haptic stimulators 214a or 214b of FIG. 2A or 2B.

[0112] With reference to FIGS. 1 and 6, in various embodiments and as mentioned above with reference to FIG. 1, the sensory augmentation system 110 comprises one or more visual devices 26. In various embodiments, the one or more visual devices 26 may be configured to project images that correspond meaningfully to the haptic stimulation provided by the smart ring 112.

[0113] The visual device 26 may provide further functions. For instance, the visual device 26 may provide any visual input corresponding to the haptic input or complementary thereto. The visual device 26 may also provide unrelated input. For example, in various instances, the visual device 26 may provide for augmented reality experiences. The visual device 26 may provide for overlaid textures on a surrounding environment of a user. For example, the visual device 26 may provide for projection of images corresponding to textures that overlay perceived real world items. For instance, a smooth surface may, when looked upon by a user of visual device 26, be made by visual device 26 to appear rough. The visual device 26 may be configured to provide various visual augmented reality experiences to supplement or augment haptic stimulation provided by the smart ring 112 to further communicate related information to a user.

[0114] Referencing FIG. 7, a method 700 of sensory augmentation is provided. The method may include providing, by a computing device, a wireless transmission corresponding to text (block 702). The method may include receiving by a first electro-haptic stimulator the wireless transmission (block 704). The method may further include generating, by a signal generation and sensing processor of the first electro-haptic stimulator, a first instruction to an electrode driver, the first instruction corresponding to a first stimulation waveform representing the text (block 706). The method may include generating, by the electrode driver the first stimulation waveform in response to the first instruction

(block 708), and providing, by an electrode array driven by the electrode driver, electrical stimulation to a human body, the electrical stimulation corresponding to the text (block 710). In various embodiments, the method further includes projecting, by glasses having a built-in image projector, an image corresponding to the text (block 712).

[0115] Referencing FIG. 8, and with additional reference to FIGS. 1-7, a use scenario 800 using multiple aspects of the sensory augmentation platform 100 of FIG. 1 is shown. In various embodiments, the visual device 26 may further comprise a sensor 814. The sensor 814 may comprise any sensor configured to register at least one of an absolute or relative location of a smart ring 828, which may correspond to the smart ring 112 of FIG. 1, the smart ring 212a of FIG. 2A, the smart ring 212b of FIG. 2B, or the smart ring 412 of FIG. 4A. For example, the sensor 814 of the visual device 26 may comprise a camera configured to visually identify the location of the smart ring 828. The sensor 814 may comprise a depth or distance sensor, for instance, an ultrasonic sensor. The sensor 814 may comprise an electromagnetic sensor sensing non-human visible spectrum. The sensor 814 may comprise any sensor configured to determine spatial information with respect to a surrounding environment of a user. Moreover, such sensed information may be implemented to facilitate identification of an absolute or relative location of a user, a user’s head, a user’s hand, a user’s smart ring 828, and other objects. In yet further instances, the visual device 26 does not include the sensor 814, and an inertial, gyroscopic, or other sensor included in the smart ring 828 provides such data, for instance, data representative of an absolute or relative location of the smart ring 828. In either event, the sensory augmentation platform 100 is capable of determining at least one of a relative or an absolute location of the smart ring 828. Moreover in either event, the sensory augmentation platform 100 is capable of determining at least one of a relative or absolute location of objects in a surrounding environment of the user.

[0116] Additionally or alternatively, as described above with reference to the smart rings 212a and 212b of FIGS. 2A and 2B, the smart ring 828 may include one or more user sensors, such as the one or more user sensors 50a and 50b described above with reference to FIGS. 2A and 2B. In some of these examples, the position/optical sensor on the ring can determine or be used to determine ring position, gestures, and what the user is pointing at. For example, the optical sensor on the rings 212a and 212b of FIGS. 2A and 2B may be configured to measure light as described above, and transmit the one or more transmission signals 44a that include information indicating the measured light to a computing/wireless device (such as the augmentation control server 124 of FIG. 1). The augmentation control server 124 of FIG. 1 may be a spatially localized computing device that is configured to use the information indicating the measured light from each of the rings 212a and 212b of FIGS. 2A and 2B, other information (e.g., motion information from the motion sensors of the rings 212a and 212b), or both the information indicating the measured light and the other information, to determine spatial positions of the rings 212a and 212b using a triangulation algorithm. The augmentation control server 124 may determine, based on the determined spatial position of one or more of the rings 212a and 212b and a known labeled spatial datapoint corresponding to a device (e.g., a television), to determine whether one or more of the rings 212a and 212b are being pointed at the device

(e.g., the television). When the augmentation control server **124** determines that one or more of the rings **212a** and **212b** is being pointed at the device (e.g., the television), the augmentation control server **124** may treat data (such as data corresponding to the one or more transmission signals **44a** corresponding to the position and/or touch sensors **54a**, **52a** of FIG. 2A) corresponding to input (such as the input **62a** of FIG. 2A) as control instructions to control the device (e.g., the television). When the augmentation control server **125** determines to treat the data as control instructions for the device (e.g., the television), the augmentation control server **124** processes the data to determine to what control instruction(s) the input **62a** corresponds. For example, the input **62a** may correspond to touch input at a particular location on the touch sensor **52a** of the ring **52a**, the augmentation control server **124** may receive the one or more transmission signals **44a** that indicate the particular location, and the augmentation control server **124** may determine that the particular location corresponds to increasing a volume on the device (e.g., a television). The augmentation control server **124** may then send an instruction to the device (e.g., a television) to cause the television to increase the volume. Additionally or alternatively, the pointing information may be displayed on a device screen, thus the ring can act as a virtual pen and even turn a non-touchscreen device into a touchscreen device. Additionally or alternatively, as described above with reference to the smart rings **212a** and **212b** of FIGS. 2A and 2B, the smart ring **828** may be configured to detect swipes or other touch input and communicates the touch input to another device.

[0117] Consequently, the sensory augmentation platform **100** is able to ascertain hand movements by a user wearing the apparatus on a finger or otherwise ascertain the positioning by the user of the smart ring **828**. Thus, the sensory augmentation platform **100** of FIG. 1 may be configured to determine that a user is pointing, gesturing, or otherwise, through movement of the smart ring **828**, providing relevant information input. In various instances, a user may point a hand with a smart ring **828** at a device desired to be controlled or otherwise interacted with via the sensory augmentation platform **100**. For example, a user may point at a television **801** to turn the television on or off, or may gesture up or down or left or right to control channel and/or volume of the television **801**. Moreover, the user may point or otherwise gesture to control many devices, or to query for a status of the device. For example a user may point or otherwise gesture to control a game system **804**, a smart home hub **802**, an electronic lock **803** and/or the like. In various instances, the visual device **26** displays augmented reality data corresponding to a status of the television **801**, game system **804**, smart home hub **802**, and/or electronic lock **803**. In further instances, an application operating on a handheld computing device **120** depicts similar status information and/or permits control of the indicated devices. The visual device **26** may provide a television status indication **805** indicating that television **801** is on, and a game system indication **808** indicating that the game system **804** is off. Moreover, a lock status indication **809** may be depicted to allow ready determination of a locked or unlocked status of an electronic lock **803**.

[0118] Furthermore, control aspects for devices may include both visual depictions via the visual device **26** and machine-human interactions by the smart ring **828**. For example, the smart home hub **802** is shown playing music

806 and an option to skip a song is shown **807**. A user may point, or otherwise interact with a point in space, such interaction detectable by the sensor **814** and/or by the smart ring **828**, the point in space corresponding to a point in space overlaid by the projected text on the visual device **26** offering the option to skip **807**. In response to a user pointing or touching such a point in space, the sensory augmentation system **100** may interoperate with the smart home hub **802**, directing it to take a responsive action such as to skip a song being played, etc.

[0119] Thus, appreciating the teachings herein above and with reference to FIGS. 1-8 various use cases are possible. For instance, as mentioned, one or more sensor **814** on one or more visual device **26** facilitates generation of a three-dimensional model of that which a user is looking at. Moreover, the smart ring **828** may be communicatively coupled, directly or indirectly, to the visual device **26** or otherwise logically associated therewith. Consequently, a simulated reality may be modeled. The modeled simulated reality may comprise a three dimensional representation of a surrounding environment of a user and movements of a user and/or a user's smart ring **828** within the surrounding environment. In various embodiments, this modeled simulated reality may be electronically combined with further modeled simulated realities of further users of further smart ring **828** and/or visual devices **26**.

[0120] Consequently, not only may a user interact with aspects of a surrounding environment via a visual device **26** and/or the smart ring **828**, but a user may interact with aspects of a surrounding environment of a different user, via a visual device **26** and/or the smart ring **828**. For instance, a visual device **26** may project images corresponding to a surrounding environment of a different user, or a simulated surrounding environment that is virtual in nature, thereby facilitating interaction with virtual realities. For instance, an architect and a customer may each wear a visual device **26** and one or smart ring **828**. The architect may invite the customer to enter a simulated surrounding environment comprising an electronic model of a proposed building designed by the architect, or may invite the customer to interact with simulated items such as architectural plans and/or the like projected on the visual device **26**.

[0121] A user may interact with a simulated environment or an environment of a different user via gestures or inputs measured by the smart ring **828** using the one or more user sensors **50a** or **50b** described above with reference to FIG. 2. Moreover, once such an environment is mapped by the visual device **26**, a user may continue to interact with the environment via the smart ring **828** even without wearing the visual device **26**. For instance, the smart ring **828** may include the one or more user sensors **50a** or **50b**, or may connect to one or more sensors of a handheld computing device **120** in order to determine its position within the environment. For instance, a user may point at a television **801** and turn the television on or off even without the visual device **26** to depict a status **805** of the television **801**. Thus, one may appreciate that the smart ring **828** includes a wide variety of user sensors **50a** or **50b** of FIGS. 2A and 2B and the visual device **26** may include a wide variety of sensors **814**. Sensors **814** may include accelerometers, gyroscopes, GPS, compasses, LIDAR, sonar, RADAR, and/or the like. Moreover, a separate illuminator device may be utilized to paint a constellation or pattern on an environment, including a pattern of light invisible to a human, which may be sensed

by sensor **814** to ascertain spatial information regarding a location, such as a location of the smart ring **828** in an environment.

[0122] Thus, the sensory augmentation platform **100** may sense a user's surrounding environment, create a three-dimensional model of an environment, and permit interaction with the model via haptic stimulators, such as the haptic stimulators **114** of FIG. 1. The model may be shared with other users and may be projected onto other visual devices **26** so that other users may, utilizing their own smart ring **828**, remotely interact with the model. Furthermore, a user may generate a unique user profile, so that the motion of the user and the user's smart ring **828** maybe specifically, accurately, and precisely tailored to the biology and personal attributes of the user. For instance, a user may move one's finger having a smart ring **828** through a full range of user arm motion to plot the extreme boundaries of potential motion. Such motion may be plotted relative to a location of a visual device **26** which may be worn on the user's head. In this manner, both the visual device **26** and the smart ring **828** may be calibrated for use by a specific user, and such calibration stored in a unique user profile.

[0123] While a computing device **120** has been discussed, as has an augmentation control server **124**, in various instances, one or more additional computing devices **120** may be provided, called a local network node. The local network node may comprise a location specific processing module reposed within a specific physical site. In further instances, the local network node may be a logical instance running remotely within an augmentation control server **124** and accessible such as by a network. The local network node may be associated with a location, a user, an account and/or a combination thereof. In various embodiments, access to the local network node requires recurring subscription payments. In various instances, the three-dimensional model of the environment may be processed and/or stored in the local network node. In this manner, processor and memory included on the smart ring **828** and/or visual devices **26** may be ameliorated.

[0124] Moreover, the local network node may be implemented to facilitate additional features of a three-dimensional model of an environment. For instance, a user may create a home area, map the home area, assign locations to controllable devices in the home area, and automatically gain access to control the controllable devices upon entry into the home area. For instance, a local network node may be associated with a living room and may facilitate control of a television by pointing with the smart ring **828** in the direction of the television. Similarly, a local network node may facilitate control of a "virtual white board" whereby a user may draw in space on a blank wall of a room, or may load textures onto surfaces, the drawings and textures stored and at runtime, rendered, by the local network node.

[0125] In yet further use cases, a store may implement a local network node. A store may map products and store this mapping in a local network node, such as by walking about the room with a visual device **26** and/or with the smart ring **828**. Subsequently, a customer may enter the store and make purchases by picking up an article, permitting the visual device **26** and/or the smart ring **828** to determine the selection of the object in connection with a local network node, and may facilitate purchasing of the item by an interaction with the smart ring **828**. Thus, one may appreciate that a local network node may be locked to one or more

user, or may permit connection by new users, such as customers, who enter a geographic region associated with the perimeter of a store.

[0126] Yet furthermore, a local network node may facilitate prompt alerting of police, fire, rescue services, and other resources as desired. For example, a local network node may facilitate calling the police in connection with detecting a pattern of user behavior by visual devices **26** or the smart ring **828** corresponding to a fall by a specific elderly user, but not when the fall corresponds to a user who is a gymnast. Alternatively, the local network node may facilitate calling the police in connection with detecting a pattern of user behavior by visual devices **26** and/or the smart ring **828** of any user corresponding to a fall and then a particular elapsed delay in arising from the fall, or in connection with pointing and/or gesturing by the user at a specific panic button or the like.

[0127] Yet furthermore, additional plug-in services may be implemented, whether on an as-needed basis, or a subscription basis. For instance, a user may activate a GPS mapping plug-in automatically upon connection to a local network node associated with an automobile, thereby facilitating provision of turn-by-turn instructions through haptic stimulation or visual indication upon entry into the vehicle, while not burdening the smart ring **828** or the visual device **26** with the memory and computational loads associated with GPS mapping when not in the vehicle.

[0128] FIG. 9 illustrates a method **900** implemented by the smart ring described herein. For example, the method **900** may be implemented using the smart ring **112** of FIG. 1, **212a** or **212b** of FIGS. 2A and 2B, **412a** or **412b** of FIG. 4A, or **512** of FIG. 5. The method **900** includes, at **910**, detecting, using one or more sensors of the smart ring, movement information regarding a finger of a user of the apparatus. For example, the movement information may correspond to the information **62a** of FIG. 2A, and the one or more user sensors may correspond to the position sensor **54a**. The method **900** further includes, at **912**, transmitting data based on the movement information to a second device. For example, the data may correspond to the one or more transmission signals **44a** provided by the processor **218a** and transmitted by the communications interface **227a**. In some aspects of the method **900**, the apparatus comprises a ring. In some aspects of the method **900**, the one or more user sensors comprise a position sensor (such as the position sensor **54a** of FIG. 2A), and the movement information corresponds to human readable message elements or to control instructions regarding the apparatus or another device as described above with reference to FIG. 2A. In at least some examples, the detection at **910** and/or the transmission at **912** of the movement information is performed responsive to the smart ring receiving an input that causes the smart ring to be in an active operational state. In other examples, steps **910** and/or **912** are not performed, despite movement of a user's finger wearing the smart ring, responsive to the smart ring being in an inactive operational state (e.g., not receiving an input to cause the smart ring to be in the active operational state).

[0129] FIG. 10 illustrates a method **1000** implemented by a smart ring described herein. For example, the method **1000** may be implemented using the smart ring **112** of FIG. 1, **212a** or **212b** of FIGS. 2A and 2B, **412a** or **412b** of FIG. 4A, or **512** of FIG. 5. The method **1000** includes, at **1010**, receiving data indicating a near field frequency associated

with access permission for accessing a second device. For example, the data may correspond to the data 72a of FIG. 2 and may be received by the network interface 227a of FIG. 2 as described above with reference to FIG. 2A. The method 1000 further includes, at 1012, enabling accessing the second device by transmitting a signal corresponding to the near field frequency. For example, the signal corresponding to the near field frequency may correspond to the signal 63a described above with reference to FIG. 2. In some aspects of the method 1000, the device comprises an automated lock or a vehicle, and accessing the device comprises unlocking the automated lock or starting the vehicle as described above with reference to FIG. 2A.

[0130] FIG. 11 illustrates a method 1100 implemented by a smart ring described herein. For example, the method 1100 may be implemented using the smart ring 112 of FIG. 1, 212a or 212b of FIGS. 2A and 2B, 412a or 412b of FIG. 4A, or 512 of FIG. 5. The method 1100 includes, at 1110, obtaining, using a ring worn on a finger of a user, position data of the ring, wherein the position data comprises a position of the ring and any changes in the position of the ring. The method 1100 further includes, at 1112, sending the position data to another device. In some aspects of the method 1100, the position data identifies a movement by a hand of the user, and the position data sent to the other device directs an action on the other device. In some aspects of the method 1100, the movement is pointing, and the ring comprises an optical sensor oriented outward from the ring towards an end of the finger. In some of these examples, the optical sensor is configured to detect an item to which the user is pointing, and the method 1100 further includes sending an indication of the item to the other device. In some aspects of the method 1100, the movement is swiping, and the ring comprises an optical sensor oriented outward from the ring towards an end of the finger. In some examples, the optical sensor is configured to detect an item to which the user is swiping, and the method 1100 further includes sending an indication of the item to the other device. In at least some examples, the obtaining at 1110 and/or the sending at 1112 of the position data is performed responsive to the smart ring receiving an input that causes the smart ring to be in an active operational state. In other examples, steps 1110 and/or 1112 are not performed, despite changes in position of the smart ring, responsive to the smart ring being in an inactive operational state (e.g., not receiving an input to cause the smart ring to be in the active operational state).

[0131] Having discussed aspects of a system and method provided herein, general features relevant to electronic communication are further outlined below. As used herein, the term “network,” such as with respect to a network which may comprise at least a portion of the network 122 illustrated in FIG. 1, includes any cloud, cloud computing system or electronic communications system or method which incorporates hardware and/or software components. Communication among the parties may be accomplished through any suitable communication channels, such as, for example, a telephone network, an extranet, an intranet, Internet, point of interaction device (point of sale device, personal digital assistant (e.g., iPhone®, Palm Pilot®, Blackberry®), cellular phone, kiosk, etc.), online communications, satellite communications, off-line communications, wireless communications, transponder communications, local area network (LAN), wide area network (WAN), virtual private network (VPN), networked or linked devices, keyboard,

mouse and/or any suitable communication or data input modality. Moreover, although the system is frequently described herein as being implemented with transmission control protocol/Internet protocol (TCP/IP) communications protocols, the system may also be implemented using Inter-network Packet Exchange (IPX), Appletalk, IP-6, Network Basic Input/Output System (NetBIOS), Open Systems Interconnection (OSI), any tunneling protocol (e.g. Internet Protocol Security (IPsec), Secure Shell (SSH)), or any number of existing or future protocols. If the network is in the nature of a public network, such as the Internet, it may be advantageous to presume the network to be insecure and open to eavesdroppers. Specific information related to the protocols, standards, and application software utilized in connection with the Internet is generally known to those skilled in the art and, as such, need not be detailed herein. See, for example, DILIP NAIK, INTERNET STANDARDS AND PROTOCOLS (1998); JAVA 2 COMPLETE, various authors, (Sybex 1999); DEBORAH RAY AND ERIC RAY, MASTERING HTML 4.0 (1997); and LOSHIN, TCP/IP CLEARLY EXPLAINED (1997) and DAVID GOURLEY AND BRIAN TOTTY, HTTP, THE DEFINITIVE GUIDE (2002), the contents of which are hereby incorporated by reference.

[0132] A network may be unsecure. Thus, communication over the network may utilize data encryption. Encryption may be performed by way of any of the techniques now available in the art or which may become available—e.g., Twofish, Rivest-Shamir-Adleman (RSA), El Gamal, Schorr signature, digital signature algorithm (DSA), Pretty Good Privacy (PGP), Public Key Infrastructure (PKI), GNU Privacy Guard (GPG or GnuPG), and symmetric and asymmetric cryptosystems.

[0133] Any communication, transmission and/or channel discussed herein may include any system or method for delivering content (e.g. data, information, metadata, etc.), and/or the content itself. The content may be presented in any form or medium, and in various embodiments, the content may be delivered electronically and/or capable of being presented electronically. For example, a channel may comprise a website or device (e.g., Facebook, YouTube®, AppleTV®, Pandora®, xBox®, Sony® Playstation®), a uniform resource locator (“URL”), a document (e.g., a Microsoft Word® document, a Microsoft Excel® document, an Adobe.pdf document, etc.), an “ebook,” an “emagazine,” an application or microapplication (as described herein), an SMS or other type of text message, an email, Facebook, twitter, Multimedia Messaging Service (MMS) and/or other type of communication technology. In various embodiments, a channel may be hosted or provided by a data partner. In various embodiments, the distribution channel may comprise at least one of a merchant website, a social media website, affiliate or partner websites, an external vendor, a mobile device communication, social media network and/or location based service. Distribution channels may include at least one of a merchant website, a social media site, affiliate or partner websites, an external vendor, and a mobile device communication. Examples of social media sites include Facebook®, Foursquare®, Twitter®, MySpace®, LinkedIn®, and the like. Examples of affiliate or partner websites include American Express®, Visa®, Google®, and the like. Moreover, examples of mobile device communications include texting, email, and mobile applications for smartphones.

[0134] In various embodiments, the methods described herein are implemented using the various particular machines described herein. The methods described herein may be implemented using the below particular machines, and those hereinafter developed, in any suitable combination, as would be appreciated immediately by one skilled in the art. Further, as is unambiguous from this disclosure, the methods described herein may result in various transformations of certain articles.

[0135] For the sake of brevity, conventional data networking, application development and other functional aspects of the systems (and components of the individual operating components of the systems) may not be described in detail herein. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent exemplary functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in a practical system.

[0136] The various system components discussed herein may include one or more of the following: a host server or other computing systems including a processor for processing digital data; a memory coupled to the processor for storing digital data; an input digitizer coupled to the processor for inputting digital data; an application program stored in the memory and accessible by the processor for directing processing of digital data by the processor; a display device coupled to the processor and memory for displaying information derived from digital data processed by the processor; and a plurality of databases. Various databases used herein may include: client data; merchant data; utility company data; institution data; regulatory agency data; and/or like data useful in the operation of the system. As those skilled in the art will appreciate, user computer may include an operating system (e.g., Windows NT®, Windows 95/98/2000®, Windows XP®, Windows Vista®, Windows 7®, OS2, UNIX®, Linux®, Solaris®, MacOS, etc.) as well as various conventional support software and drivers typically associated with computers.

[0137] The present system or any part(s) or function(s) thereof may be implemented using hardware, software or a combination thereof and may be implemented in one or more computer systems or other processing systems. However, the manipulations performed by embodiments were often referred to in terms, such as determining or selecting, which are commonly associated with mental operations performed by a human operator. No such capability of a human operator is necessary, possible, or desirable in most cases, in any of the operations described herein. Rather, the operations may be machine operations not performable by mere human activity.

[0138] In fact, in various embodiments, the embodiments are directed toward one or more computer systems capable of carrying out the functionality described herein. The computer system includes one or more processors. The processor is connected to a communication infrastructure (e.g., a communications bus, cross over bar, or network). Various software embodiments are described in terms of this exemplary computer system. After reading this description, it will become apparent to a person skilled in the relevant art(s) how to implement various embodiments using other computer systems and/or architectures. The computer system can include a display interface that forwards graphics,

text, and other data from the communication infrastructure (or from a frame buffer not shown) for display on a display unit.

[0139] The computer system also includes a main memory, such as for example random access memory (RAM), and may also include a secondary memory. The secondary memory may include, for example, a hard disk drive and/or a removable storage drive, representing a floppy disk drive, a magnetic tape drive, an optical disk drive, etc. The removable storage drive reads from and/or writes to a removable storage unit in a well-known manner. The removable storage unit represents a floppy disk, magnetic tape, optical disk, etc., which is read by and written to by the removable storage drive. As will be appreciated, the removable storage unit includes a computer usable storage medium having stored therein computer software and/or data.

[0140] In various embodiments, secondary memory may include other similar devices for allowing computer programs or other instructions to be loaded into computer system. Such devices may include, for example, a removable storage unit and an interface. Examples of such may include a program cartridge and cartridge interface (such as that found in video game devices), a removable memory chip (such as an erasable programmable read only memory (EPROM), or programmable read only memory (PROM)) and associated socket, and other removable storage units and interfaces, which allow software and data to be transferred from the removable storage unit to computer system.

[0141] The computer system may also include a communications interface. The communications interface allows software and data to be transferred between the computer system and external devices. Examples of the communications interface may include a modem, a network interface (such as an Ethernet card), a communications port, a Personal Computer Memory Card International Association (PCMCIA) slot and card, etc. Software and data transferred via communications interface are in the form of signals which may be electronic, electromagnetic, and optical or other signals capable of being received by communications interface. These signals are provided to communications interface via a communications path (e.g., channel). This channel carries signals and may be implemented using wire, cable, fiber optics, a telephone line, a cellular link, a radio frequency (RF) link, wireless and other communications channels.

[0142] The terms “computer program medium” and “computer usable medium” and “computer readable medium” are used to generally refer to media such as removable storage drive and a hard disk installed in the hard disk drive. These computer program products provide software to the computer system.

[0143] Computer programs (also referred to as computer control logic) are stored in the main memory and/or secondary memory. Computer programs may also be received via the communications interface. Such computer programs, when executed, enable the computer system to perform the features as discussed herein. In particular, the computer programs, when executed, enable the processor to perform the features of various embodiments. Accordingly, such computer programs represent controllers of the computer system.

[0144] In various embodiments, software may be stored in a computer program product and loaded into the computer

system using a removable storage drive, hard disk drive or communications interface. The control logic (software), when executed by the processor, causes the processor to perform the functions of various embodiments as described herein. In various embodiments, hardware components such as application specific integrated circuits (ASICs) are implemented. Implementation of the hardware state machine, so as to perform the functions described herein will be apparent to persons skilled in the relevant art(s).

[0145] The various system components may be independently, separately or collectively suitably coupled to the network via data links which includes, for example, a connection to an Internet Service Provider (ISP) over the local loop as is typically used in connection with standard modem communication, cable modem, Dish Networks®, Integrated Services Digital Network (ISDN), Digital Subscriber Line (DSL), or various wireless communication methods, see, e.g., GILBERT HELD, UNDERSTANDING DATA COMMUNICATIONS (1996), which is hereby incorporated by reference. It is noted that the network may be implemented as other types of networks, such as an interactive television (ITV) network. Moreover, the system contemplates the use, sale or distribution of any goods, services or information over any network having similar functionality described herein.

[0146] “Cloud” or “Cloud computing” includes a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction. Cloud computing may include location-independent computing, whereby shared servers provide resources, software, and data to computers and other devices on demand. For more information regarding cloud computing, see the NIST’s (National Institute of Standards and Technology) definition of cloud computing at <http://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-145.pdf> (last visited August 2019), which is hereby incorporated by reference in its entirety.

[0147] As used herein, “transmit” may include sending electronic data from one system component to another over a network connection. Additionally, as used herein, “data” may include encompassing information such as commands, queries, files, data for storage, and the like in digital or any other form.

[0148] The computers discussed herein may provide a suitable website or other internet-based graphical user interface which is accessible by users. In one embodiment, the Microsoft Internet Information Server (IIS), Microsoft Transaction Server (MTS), and Microsoft SQL Server, are used in conjunction with the Microsoft operating system, Microsoft NT web server software, a Microsoft SQL Server database system, and a Microsoft Commerce Server. Additionally, components such as Access or Microsoft SQL Server, Oracle, Sybase, Informix MySQL, Interbase, etc., may be used to provide an Active Data Object (ADO) compliant database management system. In one embodiment, the Apache web server is used in conjunction with a Linux operating system, a MySQL database, and the Perl, PHP, and/or Python programming languages.

[0149] Any of the communications, inputs, storage, databases or displays discussed herein may be facilitated through a website having web pages. The term “web page” as it is

used herein is not meant to limit the type of documents and applications that might be used to interact with the user. For example, a typical website might include, in addition to standard Hypertext Markup Language (HTML) documents, various forms, Java applets, JavaScript, active server pages (ASP), common gateway interface scripts (CGI), extensible markup language (XML), dynamic HTML, cascading style sheets (CSS), AJAX (Asynchronous Javascript And XML), helper applications, plug-ins, and the like. A server may include a web service that receives a request from a web server, the request including a URL (<http://yahoo.com/stock-quotes/ge>) and an IP address (123.56.789.234). The web server retrieves the appropriate web pages and sends the data or applications for the web pages to the IP address. Web services are applications that are capable of interacting with other applications over a communications means, such as the internet. Web services are typically based on standards or protocols such as XML, Simple Object Access Protocol (SOAP), AJAX, Web Services Description Language (WSDL) and Universal Description, Discovery, and Integration (UDDI). Web services methods are well known in the art, and are covered in many standard texts. See, e.g., ALEX NGHIEM, IT WEB SERVICES: A ROADMAP FOR THE ENTERPRISE (2003), hereby incorporated by reference.

[0150] Practitioners will also appreciate that there are a number of methods for displaying data within a browser-based document. Data may be represented as standard text or within a fixed list, scrollable list, drop-down list, editable text field, fixed text field, pop-up window, and the like. Likewise, there are a number of methods available for modifying data in a web page such as, for example, free text entry using a keyboard, selection of menu items, check boxes, option boxes, and the like.

[0151] The system and method may be described herein in terms of functional block components, screen shots, optional selections and various processing steps. It should be appreciated that such functional blocks may be realized by any number of hardware and/or software components configured to perform the specified functions. For example, the system may employ various integrated circuit components, e.g., memory elements, processing elements, logic elements, look-up tables, and the like, which may carry out a variety of functions under the control of one or more microprocessors or other control devices. Similarly, the software elements of the system may be implemented with any programming or scripting language such as C, C++, C#, Java, JavaScript, VBScript, Macromedia Cold Fusion, COBOL, Microsoft Active Server Pages, assembly, PERL, PHP, awk, Python, Visual Basic, SQL Stored Procedures, PL/SQL, any UNIX shell script, and extensible markup language (XML) with the various algorithms being implemented with any combination of data structures, objects, processes, routines or other programming elements. Further, it should be noted that the system may employ any number of conventional techniques for data transmission, signaling, data processing, network control, and the like. Still further, the system could be used to detect or prevent security issues with a client-side scripting language, such as JavaScript, VBScript or the like. For a basic introduction of cryptography and network security, see any of the following references: (1) “Applied Cryptography: Protocols, Algorithms, And Source Code In C,” by Bruce Schneier, published by John Wiley & Sons (second edition, 1995); (2) “Java Cryptography” by Jona-

than Knudson, published by O'Reilly & Associates (1998); (3) "Cryptography & Network Security: Principles & Practice" by William Stallings, published by Prentice Hall; all of which are hereby incorporated by reference.

[0152] As will be appreciated by one of ordinary skill in the art, the system may be embodied as a customization of an existing system, an add-on product, a processing apparatus executing upgraded software, a standalone system, a distributed system, a method, a data processing system, a device for data processing, and/or a computer program product. Accordingly, any portion of the system or a module may take the form of a processing apparatus executing code, an internet based embodiment, an entirely hardware embodiment, or an embodiment combining aspects of the internet, software and hardware. Furthermore, the system may take the form of a computer program product on a computer-readable storage medium having computer-readable program code means embodied in the storage medium. Any suitable computer-readable storage medium may be utilized, including hard disks, CD-ROM, optical storage devices, magnetic storage devices, and/or the like.

[0153] The system and method is described herein with reference to screen shots, block diagrams and flowchart illustrations of methods, apparatus (e.g., systems), and computer program products according to various embodiments. It will be understood that each functional block of the block diagrams and the flowchart illustrations, and combinations of functional blocks in the block diagrams and flowchart illustrations, respectively, can be implemented by computer program instructions.

[0154] These computer program instructions may be loaded onto a programmable data processing apparatus to produce a machine, such that the instructions that execute on the computer or other programmable data processing apparatus create means for implementing the functions specified in the flowchart block or blocks. These computer program instructions may also be stored in a computer-readable memory that can direct a computer or other programmable data processing apparatus to function in a particular manner, such that the instructions stored in the computer-readable memory produce an article of manufacture including instruction means which implement the function specified in the flowchart block or blocks. The computer program instructions may also be loaded onto a computer or other programmable data processing apparatus to cause a series of operational steps to be performed on the computer or other programmable apparatus to produce a computer-implemented process such that the instructions which execute on the computer or other programmable apparatus provide steps for implementing the functions specified in the flowchart block or blocks.

[0155] Accordingly, functional blocks of the block diagrams and flowchart illustrations support combinations of means for performing the specified functions, combinations of steps for performing the specified functions, and program instruction means for performing the specified functions. It will also be understood that each functional block of the block diagrams and flowchart illustrations, and combinations of functional blocks in the block diagrams and flowchart illustrations, can be implemented by either special purpose hardware-based computer systems which perform the specified functions or steps, or suitable combinations of special purpose hardware and computer instructions. Further, illustrations of the process flows and the descriptions

thereof may make reference to user windows, webpages, websites, web forms, prompts, etc. Practitioners will appreciate that the illustrated steps described herein may comprise in any number of configurations including the use of windows, webpages, web forms, popup windows, prompts and the like. It should be further appreciated that the multiple steps as illustrated and described may be combined into single webpages and/or windows but have been expanded for the sake of simplicity. In other cases, steps illustrated and described as single process steps may be separated into multiple webpages and/or windows but have been combined for simplicity.

[0156] The term "non-transitory" is to be understood to remove only propagating transitory signals per se from the claim scope and does not relinquish rights to all standard computer-readable media that are not only propagating transitory signals per se. Stated another way, the meaning of the term "non-transitory computer-readable medium" and "non-transitory computer-readable storage medium" should be construed to exclude only those types of transitory computer-readable media which were found in *In Re Nuijten* to fall outside the scope of patentable subject matter under 35 U.S.C. § 101.

[0157] Systems, methods and computer program products are provided. In the detailed description herein, references to "various embodiments", "one embodiment", "an embodiment", "an example embodiment", etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. After reading the description, it will be apparent to one skilled in the relevant art(s) how to implement the disclosure in alternative embodiments.

[0158] Benefits, other advantages, and solutions to problems have been described herein with regard to specific embodiments. However, the benefits, advantages, solutions to problems, and any elements that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as critical, required, or essential features or elements of the disclosure. The scope of the disclosure is accordingly to be limited by nothing other than the appended claims, in which reference to an element in the singular is not intended to mean "one and only one" unless explicitly so stated, but rather "one or more." Moreover, where a phrase similar to "at least one of A, B, and C" or "at least one of A, B, or C" is used in the claims or specification, it is intended that the phrase be interpreted to mean that A alone may be present in an embodiment, B alone may be present in an embodiment, C alone may be present in an embodiment, or that any combination of the elements A, B and C may be present in a single embodiment; for example, A and B, A and C, B and C, or A and B and C. Although the disclosure includes a method, it is contemplated that it may be embodied as computer program instructions on a tangible computer-readable carrier, such as a magnetic or optical memory or a magnetic or optical disk. All structural, chemical, and functional equivalents to the elements of the above-

described exemplary embodiments that are known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the present claims. Moreover, it is not necessary for a device or method to address each and every problem sought to be solved by the present disclosure, for it to be encompassed by the present claims.

[0159] Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element herein is to be construed under the provisions of 35 U.S.C. 112 (f) unless the element is expressly recited using the phrase “means for.” As used herein, the terms “comprises”, “comprising”, or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

[0160] The foregoing method descriptions and the process flow diagrams are provided merely as illustrative examples and are not intended to require or imply that the steps of the various embodiments must be performed in the order presented. The steps in the foregoing embodiments may be performed in any order. Words such as “then,” “next,” etc., are not intended to limit the order of the steps; these words are simply used to guide the reader through the description of the methods. Although process flow diagrams may describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be rearranged. A process may correspond to a method, a function, a procedure, a subroutine, a subprogram, etc. When a process corresponds to a function, its termination may correspond to a return of the function to the calling function or the main function.

[0161] The various illustrative logical blocks, modules, circuits, and algorithm steps described in connection with the embodiments disclosed herein may be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present invention.

[0162] Embodiments implemented in computer software may be implemented in software, firmware, middleware, microcode, hardware description languages, or the like, or any combination thereof. A code segment or machine-executable instructions may represent a procedure, a function, a subprogram, a program, a routine, a subroutine, a module, a software package, a class, or any combination of instructions, data structures, or program statements. A code segment may be coupled to another code segment or a hardware circuit by passing and/or receiving information, data, arguments, parameters, or memory contents. Information, arguments, parameters, data, etc. may be passed, for-

warded, or transmitted via any suitable means including memory sharing, message passing, token passing, network transmission, etc.

[0163] The actual software code or specialized control hardware used to implement these systems and methods is not limiting of the invention. Thus, the operation and behavior of the systems and methods were described without reference to the specific software code being understood that software and control hardware can be designed to implement the systems and methods based on the description herein.

[0164] When implemented in software, the functions may be stored as one or more instructions or code on a non-transitory computer-readable or processor-readable storage medium. The steps of a method or algorithm disclosed herein may be embodied in a processor executable software module which may reside on a computer-readable or processor-readable storage medium. A non-transitory computer-readable or processor-readable media includes both computer storage media and tangible storage media that facilitate transfer of a computer program from one place to another. A non-transitory, processor-readable storage media may be any available media that may be accessed by a computer. By way of example, and not limitation, such non-transitory, processor-readable media may comprise RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other tangible storage medium that may be used to store desired program code in the form of instructions or data structures and that may be accessed by a computer or processor. Disk and disc, as used herein, include compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk, and Blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above should also be included within the scope of computer-readable media. Additionally, the operations of a method or algorithm may reside as one or any combination or set of codes and/or instructions on a non-transitory, processor-readable medium and/or computer-readable medium, which may be incorporated into a computer program product.

[0165] The preceding description of the disclosed embodiments is provided to enable any person skilled in the art to make or use the present invention. Various modifications to these embodiments will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other embodiments without departing from the spirit or scope of the invention. Thus, the present invention is not intended to be limited to the embodiments shown herein but is to be accorded the widest scope consistent with the following claims and the principles and novel features disclosed herein.

[0166] While various aspects and embodiments have been disclosed, other aspects and embodiments are contemplated. The various aspects and embodiments disclosed are for purposes of illustration and are not intended to be limiting, with the true scope and spirit being indicated by the following claims.

What is claimed is:

1. A smart ring, comprising:

a ring body configured to be worn on a finger of a user; at least one haptic stimulator configured to receive a drive signal and communicate data to the user via a haptic stimulation of the finger; and

- at least one processor coupled to the at least one haptic stimulator, wherein the at least one processor is configured to:
- receive a notification;
 - generate the drive signal based on the notification, wherein the drive signal has a unique correspondence to a characteristic of the notification and corresponds to a temporal vibration pattern; and
 - drive the at least one haptic stimulator according to the drive signal to communicate the data to the user, the data being representative of the characteristic.
2. The smart ring of claim 1, wherein the notification is one of text data, an alert, or a social media notification.
3. The smart ring of claim 1, wherein the characteristic of the notification comprises a content of the notification or a type of the notification.
4. The smart ring of claim 1, wherein the notification is a first notification, the characteristic is a first characteristic, and the temporal vibration pattern is a first temporal vibration pattern, and wherein the at least one processor is configured to:
- receive a second notification;
 - generate the drive signal based on the second notification, wherein the drive signal has a unique correspondence to a second characteristic of the second notification and corresponds to a second temporal vibration pattern that differs from the first temporal vibration pattern; and
 - drive the at least one haptic stimulator according to the drive signal to communicate the data to the user, the data being representative of the second characteristic.
5. The smart ring of claim 1, wherein generating the drive signal comprises selecting the temporal vibration pattern from among multiple temporal vibration patterns based on the unique correspondence between the characteristic of the notification and the temporal vibration pattern.
6. The smart ring of claim 5, wherein each temporal vibration pattern of the multiple temporal vibration patterns uniquely corresponds to one characteristic of multiple characteristics of the notification including at least the characteristic.
7. The smart ring of claim 1, wherein the at least one haptic stimulator includes a first haptic stimulator, and wherein the temporal vibration pattern consists of multiple stimulations by the first haptic stimulator.
8. The smart ring of claim 1, wherein the at least one haptic stimulator includes a first haptic stimulator and a second haptic stimulator, and wherein the temporal vibration pattern comprises stimulation by the first haptic stimulator and the second haptic stimulator.
9. A smart ring, comprising:
- a ring body configured to be worn on a finger of a user;
 - a touch sensor coupled to the ring body, wherein the touch sensor is configured to receive a touch input from the user and convert the touch input into an input signal;
 - a motion sensor coupled to the ring body, wherein the motion sensor is configured to provide position data representative of a position of the smart ring in a Cartesian coordinate system;
- at least one processor coupled to the capacitive touch sensor, wherein the at least one processor is configured to:
- receive the input signal;
 - responsive to receiving the input signal:
 - activate the motion sensor; and
 - obtain the position data;
- form a transmission signal based on the position data; and
 - a communications interface coupled to the at least one processor, wherein the communications interface is configured to transmit the transmission signal to another device.
10. The smart ring of claim 9, wherein the motion sensor detects motion in six degrees of freedom.
11. The smart ring of claim 9, wherein the position data is provided in three dimensions.
12. The smart ring of claim 9, wherein the transmission signal comprises the position data.
13. The smart ring of claim 9, wherein the position information is determined with respect to the another device.
14. A method, comprising:
- receiving a data input;
 - mapping the data input to a temporal vibration pattern from among multiple temporal vibration patterns; and
 - controlling a haptic stimulator to provide haptic stimulation according to the temporal vibration pattern.
15. The method of claim 14, further comprising:
- detecting, via a touch sensor, a gesture; and
 - transmitting a control signal to a device to control the device according to the gesture.
16. The method of claim 15, wherein the gesture is a swipe gesture, and wherein the control signal comprises a scroll command to cause the device to perform a scroll operation responsive to the control signal.
17. The method of claim 14, wherein the data input is one of text data, an alert, or a social media notification.
18. The method of claim 14, wherein each type of data input uniquely corresponds to one of the multiple temporal vibration patterns.
19. The method of claim 14, further comprising:
- receiving a second data input;
 - mapping the second data input to a second temporal vibration pattern from among the multiple temporal vibration patterns; and
 - controlling the haptic stimulator to provide second haptic stimulation according to the second temporal vibration pattern.
20. The method of claim 14, wherein the data input is mapped to the temporal vibration pattern based on a content of the data input or a type of the data input.
21. A computer-program product comprising computer-executable instructions stored on at least one non-transitory computer medium, the instructions executable by at least one processor to cause the at least one processor to:
- receive a notification;
 - generate a drive signal based on the notification, wherein the drive signal corresponds to a vibration pattern and has a one-to-one correspondence to a characteristic of the notification; and
 - drive at least one haptic stimulator according to the drive signal to convey the data to the user, the data being representative of the characteristic.
22. The computer-program product of claim 21, wherein the notification is one of text data, an alert, or a social media notification, and wherein the characteristic of the notification comprises a content of the notification or a type of the notification.
23. The computer-program product of claim 21, wherein generating the drive signal comprises selecting the

vibration pattern from among multiple vibration patterns based on the one-to-one correspondence between the characteristic of the notification and the vibration pattern.

24. The computer computer-program product of claim **21**, wherein the at least one haptic stimulator includes a first haptic stimulator, and wherein the vibration pattern consists of multiple stimulations by the first haptic stimulator.

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